Digital Fault Recorder for High Impedance Faults

Submitted in partial fulfillment of the requirements for the degree of

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in

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by

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May 2026

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We hereby declare that the thesis entitled *"**Digital Fault Recorder for High Impedance Faults**"* submitted by **Vijay Kumar Meena**, **Kaumudi Sharma**, and **Kirtesh Singh**, for the award of the degree of Bachelor of Technology in Electrical and Electronics Engineering to VIT University is a record of bonafide work carried out by us under the supervision of **Dr. Himadri Lala**, Assistant Professor, School of Electrical Engineering, VIT University, Vellore.

We further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

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Kirtesh Singh
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This is to certify that the thesis entitled *“**Digital Fault Recorder for High Impedance Faults**”* submitted by **Vijay Kumar Meena** (22BEE0348), **Kaumudi Sharma** (22BEE0368), and **Kirtesh Singh** (22BEE0344), School of Electrical Engineering, VIT, Vellore, for the award of the degree of Bachelor of Technology in Electrical and Electronics Engineering, is a record of bonafide work carried out by them under my supervision during the period 05.12.2025 to 21.04.2026, as per the VIT Code of Academic and Research Ethics.

The contents of this report have not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The thesis fulfills the requirements and regulations of the University and, in my opinion, meets the necessary standards for submission.

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The thesis is satisfactory / unsatisfactory

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Head of the Department

Dean, SELECT EEE

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I wish to express my profound gratitude to my parents for their unconditional support, patience, and encouragement throughout my academic journey.

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EXECUTIVE SUMMARY

High impedance faults (HIF) in power lines usually evade overcurrent relays due to low currents, which might fall to even below ten percent of normal operating conditions. An active conductor coming into contact with a conductor can cause arcing, thereby giving rise to unique waveform characteristics. To detect such HIF events, a detection function routine (DFR) has been designed. The experimental investigation was conducted at the High Voltage Laboratory, VIT University, using a high voltage transformer with a leaning-tree and sand model setup. Real-time voltage readings are captured using an Arduino board coupled with the ZMPT101B voltage sensor. Analysis and evaluation of signals were done using a computer running Python code interfaced with the physical model. In the leaning-tree experiment, the voltage waveform generated by arcing exhibited distinctive third-harmonic signatures at 150 Hz, which is characteristic of a high-impedance fault. The system detected the fault immediately after its occurrence, demonstrating that low-cost hardware equipment valued below 800 rupees, along with free software, could provide reliable results. Despite its rudimentary design, the model was capable of instant detection of small electrical changes. However, the most remarkable feature was the ability of uncomplicated hardware to recognize intricate signal patterns without additional tuning despite environmental noise.

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LIST OF TERMS AND ABBREVIATIONS

DFR Digital Fault Recorder. 2, 20, 51, 54

FFT Fast Fourier Transform. 8

HIAF High Impedance Arc Fault. 2, 54

HIF High Impedance Fault. 2, 5, 17, 20

PSD Power Spectral Density. 8, 17

CHAPTER 1

INTRODUCTION

High impedance arc faults are difficult to detect with traditional protective measures. The conventional method is based on abrupt changes in the current level. However, high impedance arc faults seek to minimize such abrupt current changes in order to escape detection. In addition, the flow rate is typically less than ten percent of the usual rate. However, as the fault occurs, heat will gradually increase, reaching several thousands of degrees centigrade given enough time. Such discrepancies are quite typical in practice.

Since trees always lean in certain ways, the nature of the arcs created by them will be different from other pieces of equipment since fault current takes a path dependent upon the grounding resistance of the equipment. What is more remarkable about high impedance arc faults is the impact that the surface of contact has upon the process of analysis. Surface conditions affect both arc impedance and conductivity, which leads to unpredictable responses from a tree while arcing.

Intense heat emanates rapidly from the charged electrical device, followed by pressure waves due to quick expansion. Arc flashes may be caused by improper connections between live conductors and earth-grounded objects such as metals and trees. In some instances, the electricity will arc through space before reaching the ground. This is because there exists great voltage stress in that space compared to its small size. As the insulation gets breached, the electricity arcing occurs.

Midway through the paragraph, the decrease in moisture in the system alters the resistance of the current flow in the trees. Despite the continuity of the electricity flow, the properties of the substance through which it flows are not static and unpredictable. It does not mean all the changes must recur in any predictable fashion, which can happen at any moment. The resistance of the current increases intermittently due to the change in the substance. Dynamic arcs appear whenever there exist flickering contact points in the changing load. The patterns disintegrate rapidly since there is no consistency in bridging the voltage gap. Impedance determines what people consider a static arc.

1.1 Objective

The intended output for the project is to develop an economical DFR that will be able to detect HIF in power networks. Given the fact that the fault current is extremely small, often less than 10% of the actual current load, it merges with ordinary events in the system. The problem arises since normal overload relays cannot differentiate the weak fault current in such a case from ordinary functioning in the network. [1, 2]

An Arduino-based DFR utilizes the use of a chip to collect actual signals in a high-voltage test set up, with Python being used to refine the data and break down frequencies. The objectives of the research are to ensure that the following is accomplished:

- To set up a tree-leaning arc fault experiment in the VIT High Voltage Engineering Laboratory.
- To acquire arc voltage signals in real time using an Arduino microcontroller and ZMPT101B voltage sensor based data acquisition system.
- To analyze the acquired arc voltage signals in Python and extract their harmonic signatures.
- Identify the predominant frequency components of High Impedance Arc Fault (HIAF) and distinguish tree-leaning arc faults from normal operating conditions.
- To validate the system as a practical and cost-effective Digital Fault Recorder (DFR) for High Impedance Fault (HIF) detection.

1.2 Motivation

The touch of a live wire on a tree will create an arc which is hot enough to cause fires, and that describes HIF. As the electric current involved in this phenomenon is very low, traditional circuit breakers may not be able to detect the problem and act accordingly. In contrast to short circuits that lead to instantaneous action, the current levels are insufficient to make ordinary breakers act as usual. The risk increases because more vegetation grows around the wires, making the problem even trickier.

The first factor that explains why the research was conducted revolves around the current approach to detecting HIF by means of sophisticated devices such as expensive digital relays and special equipment that analyzes the quality of electricity. Due to the cost and difficulties associated with implementing such solutions, the purpose is to design and build a DFR system which would be simple and cheap while being effective with common equipment, like Arduino, and readily available software, such as Python.

1.3 Background

In this regard, the paper attempts to provide a reliable technique in addressing the arc signals such that it will become possible to analyze different arcing events using their harmonics.

The HIAF is sometimes experienced where the live conductor comes into contact with the earth, a tree or any other foreign object. Other times, equipment malfunctions, damaged insulation, and adjacent cables could trigger the production of an arc. High Impedance faults generally affect system performance since they remain undetected by traditional protection schemes, thereby reducing reliability.

An arc results from the creation of an electrical pathway through the failure of insulation to result in the breakdown of the material. The phenomenon sometimes occurs in the gas but may occur just the same way in a solid or liquid medium, especially where insulation becomes weakened.

Different forms of arcs could exist depending on environmental conditions and the material properties. As part of analyzing the behavior of arcs at varied conditions, tests have been conducted on situations where arcing occurs in soil or wet sand.

In this case, the attention will be given to arcs resulting from tilted trees located on a medium voltage power cable. Several characteristics will be considered during this analysis.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 Review of Literature

Despite being challenging to detect, High Impedance Faults cause problems in electrical networks due to the lack of sufficient currents to trigger overcurrent protection units. Early research done by Tengdin et al. [1] and others found that the arcs' behavior is erratic; the flashing is what characterizes this type of fault. Later, Aucoin and Jones [2] concluded that an unpredictable arc was the fundamental characteristic. The reason why this failure is difficult to detect is mainly its pulsed nature with no continuous flow.

Various techniques could be used to analyze the signal characteristics. Power Line Communication (PLC), previously discussed in [3], could be employed to determine anomalies. Shape-based mathematical signal processing techniques, as mentioned in [4], can assist in detection too. Nonetheless, analyzing harmonics is considered more efficient since electric arcs cause current distortion in specific frequency bands [6]. Although modern research concentrates on deep neural networks or wavelets analysis [9], they require substantial computing power.

In recent years, practical approaches use inexpensive microcontrollers to monitor the system in real-time. The frequencies' evaluation through FFT and PSD reveals distinct odd harmonics (3rd, 5th, and 7th) associated with the electric arcs. Based on this idea of frequency detection, this project will utilize a simplified detection algorithm within a common Arduino-Python framework.

2.2 Project Description

The output in this case is the design of a custom-made Digital Fault Recorder (DFR). The design takes a combination of hardware and software and combines them to create a seamless flow of work. Data capturing occurs instantly, with no lag. Processing is instantaneous; it breaks frequencies instantly. It occurs seamlessly as each part goes into another instantly. Each part in real time operation has specific behavior. Real-time operation produces a natural occurrence of spectral analysis.

Input in this instance is provided by a one-phase power line simulator and directed to a ZMPT101B step down transformer. Since it provides isolation of electric signals, it lowers the voltages to a level ranging from zero to five volts that are safe for the microcontroller. The Arduino reads and captures these voltage changes as data streams at intervals, producing numbers for each value of current in time. Each value then goes through a serial interface in a UART protocol to a computer system connected for storing and analyzing data.

Today, Python is the preferred tool for performing most processes in digital signal processing. Signals captured in time are transformed to a frequency domain using DFT and PSD. In doing so, rather than looking at the entire wave form, it analyzes each band frequency contribution. Rather than showing smooth curves, it shows any distortion like shoulder in relation to time. Additionally, increase in odd number harmonics shows nonlinearity in the spectrum.

2.2.1 Detection Algorithm

To minimize false positives while maintaining high sensitivity, a multi-stage detection logic is implemented. A HIF is confirmed only when the following three criteria are satisfied simultaneously for ≥ 3 consecutive cycles:

1. **Spectral Trigger:** The 3rd harmonic (150 Hz) PSD contribution exceeds 5% of the total signal power, marking the onset of non-linear arcing.
2. **Temporal Trigger:** Detection of zero-crossing delays (the "shouldering" effect) or a significant rise in the dV/dt RMS value, indicating high-frequency re-ignition transients.
3. **Broadband Trigger:** Integrated energy in the 200–500 Hz noise floor band exceeds 2.5% of the total PSD energy, capturing the stochastic nature of the fault.

This multi-faceted approach effectively differentiates between genuine high-impedance ground faults and standard non-linear loads or switching transients.

2.3 Hardware and Software Requirements

2.3.1 Software

- Python 3.x: Utilized for real-time processing. Key libraries include `NumPy` for FFT computation, `SciPy.signal` for PSD estimation, and `pySerial` for data acquisition.
- Arduino IDE 2.x: Used for developing firmware to manage high-speed ADC sampling and serial data transmission.

2.3.2 Hardware

- ZMPT101B Voltage Sensor: To perform an isolated measurement of the voltage signal.
- Arduino Uno: The main data acquisition interface for analog-to-digital conversion (10-bit ADC).
- Tree-Leaning Test Setup: HV test line with the tree branch setup to mimic the HIF situation.
- Laptop: Signal analysis and visualization device.

2.4 Goals

1. Perform tree leaning arc fault experimental set up validation in the High Voltage Laboratory, VIT.
2. Design Arduino based firmware for real time collection and communication of arc fault voltages in their waveforms.
3. Design a Python based signal processing module for generation of FFT and PSD forms of the signal from the collected data.
4. Analyze the percentage contribution of each frequency to the spectrum and analyze the main odd harmonics present in the signal.
5. Test the logic design for HIF identification using Spectral, Temporal and Broadband Trigger together.
6. Develop an inexpensive HIF detector below INR 800.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 Software Specification

- **Operating System**: Windows 10/11 (64-bit).
- **Python 3.10** with: `PyEMD 1.x`, `NumPy`, `SciPy`, `Matplotlib`, `pySerial`.
- **Arduino IDE 2.x** for firmware upload.

3.2 Hardware Specification

Table 3.1 lists the key hardware components.

Table 3.1: Hardware Specification of the Digital Fault Recorder System

Sl. No.	Component	Specification
1	HV Test Transformer	High-voltage AC testing transformer, VIT HV Lab
2	Leaning-Tree Setup	Real tree branch placed near MV test line
3	Potential Divider	Scales HV signal to 0–5 V for Arduino ADC
4	Arduino Uno	10-bit ADC; max $\sim$9.6 kSa/s; UART 115200 baud
5	Laptop (Data Processor)	Intel Core i5/i7, 8 GB RAM; Python 3.10
6	USB Cable	Arduino-to-PC serial link

3.2.1 Quotation and Reference to Earlier Work

The arc fault structure, which was subjected to tests, allows us to understand harmonic behavior, relying on techniques proven to work by Lala and Karmakar [14] in the National Institute of Technology Rourkela. In the leaning tree test performed at VIT's High Voltage Engineering Lab, the devices used are arranged similar to those in the previous experiment. Unlike other experiments, in this one, EMD does not play any role, but spectral analysis helps to create a less computationally intensive environment.

3.2.2 Use of Abbreviations

All the abbreviations are written out completely the first time they are mentioned in the text; they are then referred to using their abbreviated forms thereafter. Thus, Fast Fourier Transform (FFT), which is first introduced as the Fast Fourier Transform, becomes FFT afterwards. In addition, Power Spectral Density (PSD), which stands for Power Spectral Density, is abbreviated to its initials only after its initial mention in the text.

CHAPTER 4

DESIGN APPROACH AND DETAILS

4.1 Design Approach

The customized DFR architecture operates on an unceasing stream of data inputs that follow an orderly sequence involving three distinct processes. The first one entails capturing data via hardware based on Arduino technology. Secondly is the fast transmission of the collected data via serial communications as fast as the input rate. Thirdly, a software component using Python performs frequency segmentation in order to detect patterns associated with HIFs. As opposed to analyzing pre-recorded data, the approach relies on detecting electrical variations in real time..

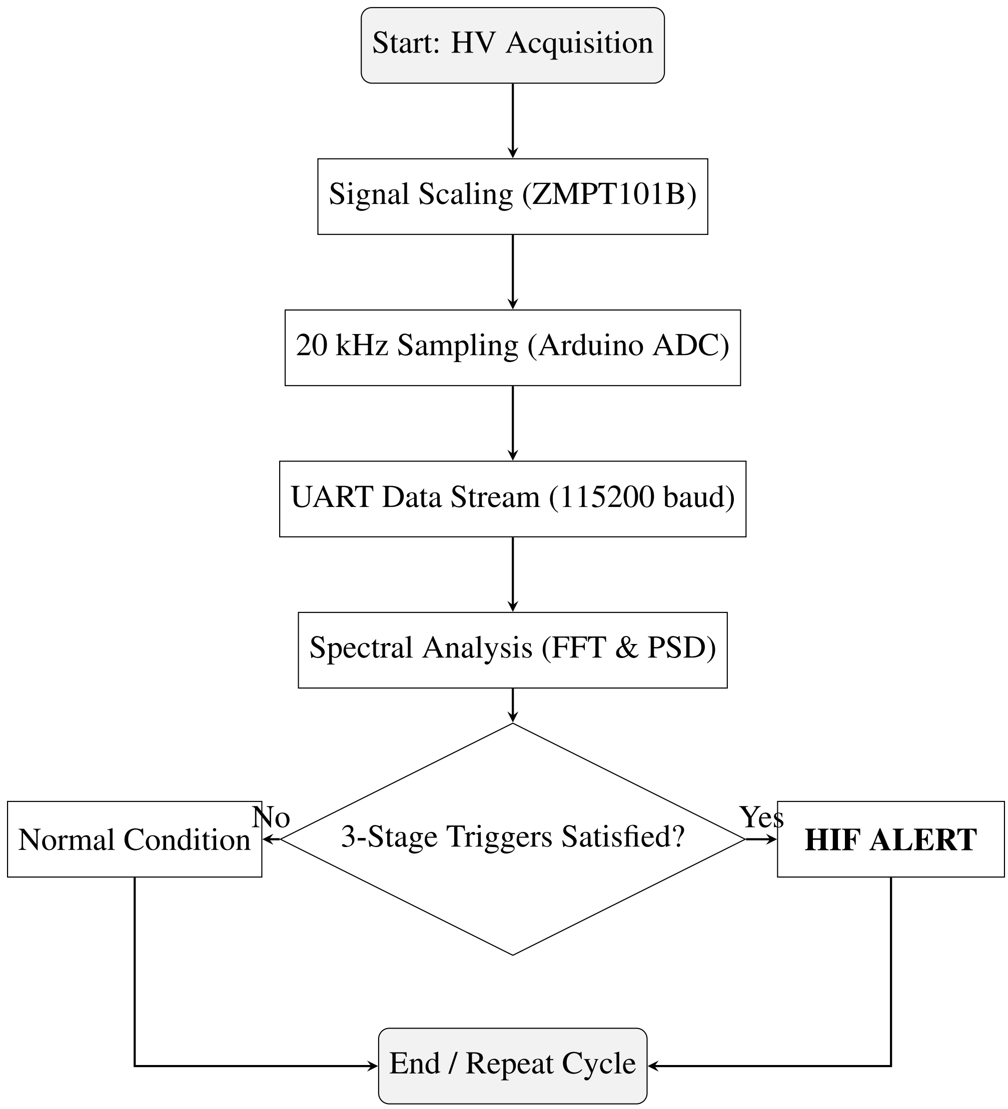


Figure 4.1: Operational Flow Chart of the DFR Detection Logic

4.2 Signal Processing Framework

Starting now with a departure from amplitude scans based on timing, the method hinges on analyzing characteristics in the frequency range. The heart of the method is the conversion of raw digital signals to frequency components.

4.2.1 Fourier Analysis and Spectral Density

The digitized signal $v[n]$ is processed using a Discrete Fourier Transform (DFT) to obtain the magnitude spectrum. To account for signal noise and power distribution, the Power Spectral Density (PSD) is calculated using the periodogram method with a Hann window to reduce spectral leakage:

$$P(f) = \frac{1}{f_s N} \left| \sum_{n=0}^{N-1} v[n]w[n]e^{-j2\pi fn/f_s} \right|^2 \tag{4.1}$$

where f_s is the sampling frequency (20 kHz), N is the number of samples, and $w[n]$ is the windowing function.

4.2.2 Harmonic Contribution and Detection Logic

This algorithm evaluates the relative contribution of each frequency component to the energy content of the signal. An instance of HIF occurs if the arc-induced non-linearities result in a specific phase deviation of the odd harmonics, especially the third harmonic (150 Hz) and the fifth harmonic (250 Hz).

The decision-making logic employs a three-stage simultaneous trigger based on the following verified thresholds:

- **Spectral Trigger:** $\mathrm{PSD}_{150\mathrm{Hz}} > 5\%$
- **Temporal Trigger:** Significant rise in dV/dt RMS indicating re-ignition transients
- **Broadband Trigger:** Integrated noise energy (200–500 Hz) $> 2.5\%$

4.3 System Implementation

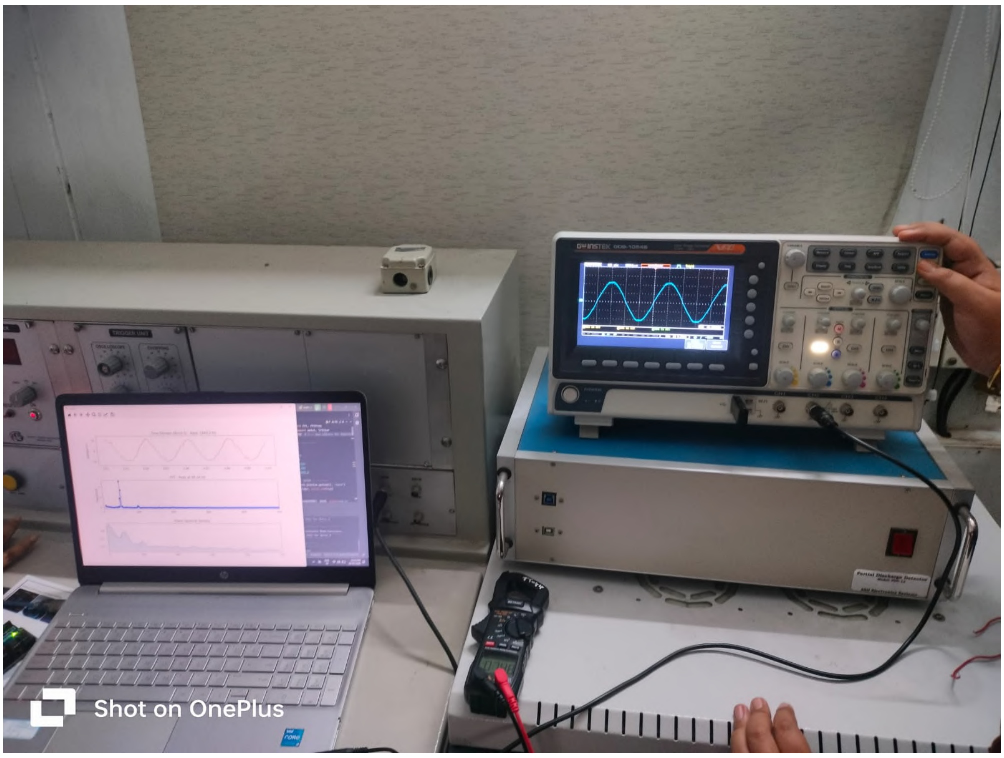


Figure 4.2: High-Voltage Equipment Arrangement

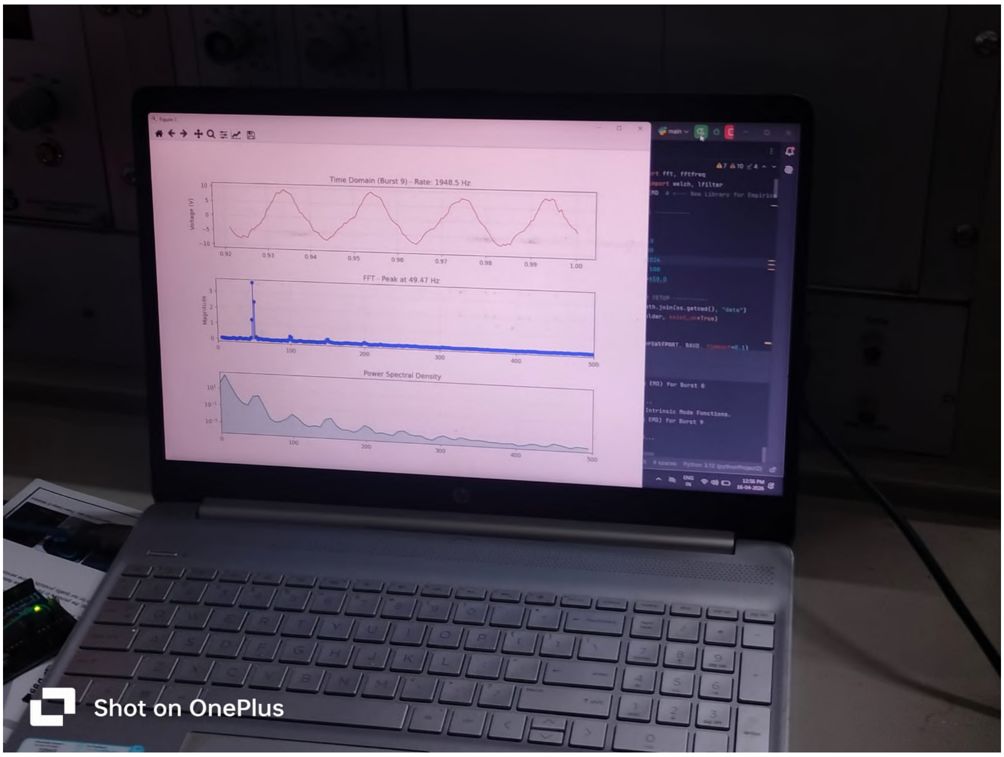


Figure 4.3: High-Voltage Equipment Arrangement

4.3.1 Arduino Signal Acquisition

The Arduino code works best at a frequency of 20 kHz. The transfer of integers using the serial port eliminates large calculations on board due to the strict time limitations set by the ADC cycles.

Listing 4.1: High-Speed Arduino Acquisition Firmware

```
// Arduino DFR - Real-time Signal Acquisition (20 kHz)
const int ANALOG_PIN = A1;
unsigned long micro_step = 50; // 50us delay for 20kHz

void setup() {
  Serial.begin(115200);
}

void loop() {
  // Continuous live streaming of ADC samples
  int val = analogRead(ANALOG_PIN);
  Serial.println(val);
  delayMicroseconds(micro_step);
}
```

4.3.2 Python Real-Time Spectral Engine

The Python engine handles the "Live Monitor" functionality. It reads the serial stream, computes the FFT/PSD, and plots the results in a 2×2 matrix for instantaneous visual verification.

Listing 4.2: Live Python Engine for Spectral and Contribution Analysis

```python
import numpy as np
from scipy import signal
import serial

def process_live_signal(data_chunk, fs=20000):
    # Detrend and normalize
    sig = data_chunk - np.mean(data_chunk)
    n = len(sig)

    # 1. FFT Magnitude Calculation
    fft_vals = np.fft.rfft(sig)
    fft_mag = np.abs(fft_vals) * (2.0 / n)
    freqs = np.fft.rfftfreq(n, d=1/fs)

    # 2. PSD Calculation
    f_psd, p_psd = signal.periodogram(sig, fs, window='hann')

    # 3. Harmonic Contribution Analysis
    total_mag = np.sum(fft_mag)
    total_pwr = np.sum(p_psd)

    # Calculate % share of specific frequency bins
    m_contrib = (fft_mag / total_mag) * 100
    p_contrib = (p_psd / total_pwr) * 100

    return freqs, fft_mag, f_psd, p_psd, m_contrib, p_contrib

# Execution logic:
# Read from serial -> process_live_signal -> Display UI
```

4.4 Codes and Standards

The implementation adheres to the following industry guidelines for power quality and harmonic monitoring:

- **IEEE Std 1159-2019**: Recommended Practice for Monitoring Electric Power Quality.

- **IEEE Std 519-2014**: IEEE Standard for Harmonic Control in Electric Power Systems.
- **IEEE Std C37.111-2013**: Standard for Common Format for Transient Data Exchange (COMTRADE).
- **IEC 60255-151**: Requirements for overcurrent/undercurrent protection.

4.5 Constraints, Alternatives and Tradeoffs

4.5.1 Constraints

- **ADC Resolution**:The ADC on the Arduino Uno, which operates on a 10-bit architecture, limits the voltage resolution to $\sim$4.88 mV/bit. Though sufficient for high-energy events, the precise determination of low-amplitude harmonic distortion calls for an accurate calibration of the ZMPT101B's gain.
- **Processing Latency**: The real-time FFT and PSD computations performed using Python will have some delay due to buffering operations. To maintain the real-time aspect, it becomes imperative to optimize the window length.
- **Sensor Bandwidth**: The bandwidth of the voltage transformer ZMPT101B has been measured to have linearity in response up to extremely high frequencies; however, it is still ideal for HIF operation within the 50–500 Hz range.
- **Safety and Environment**: It is crucial to strictly adhere to all safety measures while conducting any experiment involving high voltage in the VIT HV Laboratory.

4.5.2 Alternatives

- **Standalone DSP/FPGA**: Has the advantage of being able to process data fast and sample at high rates, but is much more expensive and difficult to program.
- **Commercial Power Quality Analyzers**: Very accurate, but work like "black boxes," and cannot support the novel three-step triggering algorithm used here.
- **Wavelet Transform (DWT)**: Useful for analyzing transients, but it needs mother wavelets to be defined, and its computation would be more demanding than that of the proposed FFT-based spectral contribution method.

4.5.3 Tradeoffs

- **Sampling Rate vs. Accuracy**: Increasing the sampling rate of the Arduino to 20 kHz ensures that higher order harmonics and transients can be sampled accurately. This would be the drawback in terms of data being transmitted at a higher speed via UART; however, the problem could be resolved by ensuring that the baud rate used is 115,200.

- **Cost vs. Performance**:Using the Arduino with ZMPT101B configuration (total cost around INR 800) provides around a hundred times reduction in cost compared to the NI-DAQ setup. The disadvantage in terms of performance is almost negligible because the signatures of interest (the 3rd and 5th harmonics) are still well below the 10 kHz Nyquist threshold.

- **Odd vs. Even Harmonic Focus**: While some references might suggest even harmonics for detecting fault current, this study focuses on odd harmonic components (150 Hz, 250 Hz). This ensures more robust detection in the symmetric arc pattern of tree leaning faults.

CHAPTER 5

SCHEDULE, TASKS AND MILESTONES

5.1 Schedule

During the span of a single semester from December 2025 to May 2026, the entire project followed a systematic sequence of events. First of all, a detailed literature survey was performed alongside the calibration of the sensors. Following this step, the integration of hardware and firmware took place so that there could be quick data generation. While the steps may not have been sequential in nature, the process of developing a spectral analyzer using Python went on alongside the testings. In the high voltage laboratory, tests confirmed the functionality of the apparatus.

5.2 Tasks and Milestones

Table 5.1: Project Tasks and Milestones

No.	Task	Duration	Milestone
1	Literature survey on HIF and spectral analysis	2 weeks	Review completed
2	Sensor calibration (ZMPT101B) and system design	1 week	Hardware architecture finalized
3	Arduino firmware development (20 kHz sampling)	2 weeks	High-speed serial acquisition ready
4	Python real-time serial interface	1 week	Live data stream established
5	FFT and PSD engine implementation	2 weeks	Spectral decomposition validated
6	Harmonic contribution and broadband noise analysis	1 week	Thresholds for 3rd harmonic set
7	HV lab experiment – leaning tree scenario	2 weeks	Arc signatures captured and live-plotted
8	Implementation of 3-stage simultaneous trigger logic	1 week	HIF detection validated
9	Results tabulation and performance analysis	1 week	System accuracy verified
10	Final report writing and formatting	3 weeks	Thesis submitted

5.3 Math Equations in Thesis

All equations are enclosed in the LaTeX environment for equations. Equations are aligned to center and are numbered sequentially in every chapter, while references to these equations use proper formatting, such as Eq. 4.1. Variables and units that involve complex quantities are expressed using mathematical notations.

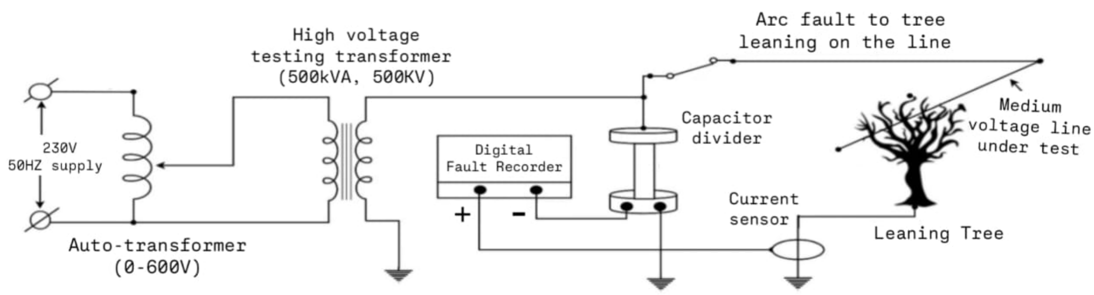
CHAPTER 6

PROJECT DEMONSTRATION

6.1 Experimental Setup

Experiments on tree proximity arcing were conducted in the VIT High Voltage Engineering Lab. In the experiment, an actual piece of tree limb was brought in proximity to a live medium-voltage power line in order to demonstrate the natural behavior of trees in terms of leaning towards the conductors in stormy weather outdoors in distribution lines. A sudden power surge moved along the circuit using a high voltage transformer, while the magnitude of the current grew gradually in a controlled manner. The phenomenon of arcing started when there was enough electrical stress on the gap between the plant and the conductor.

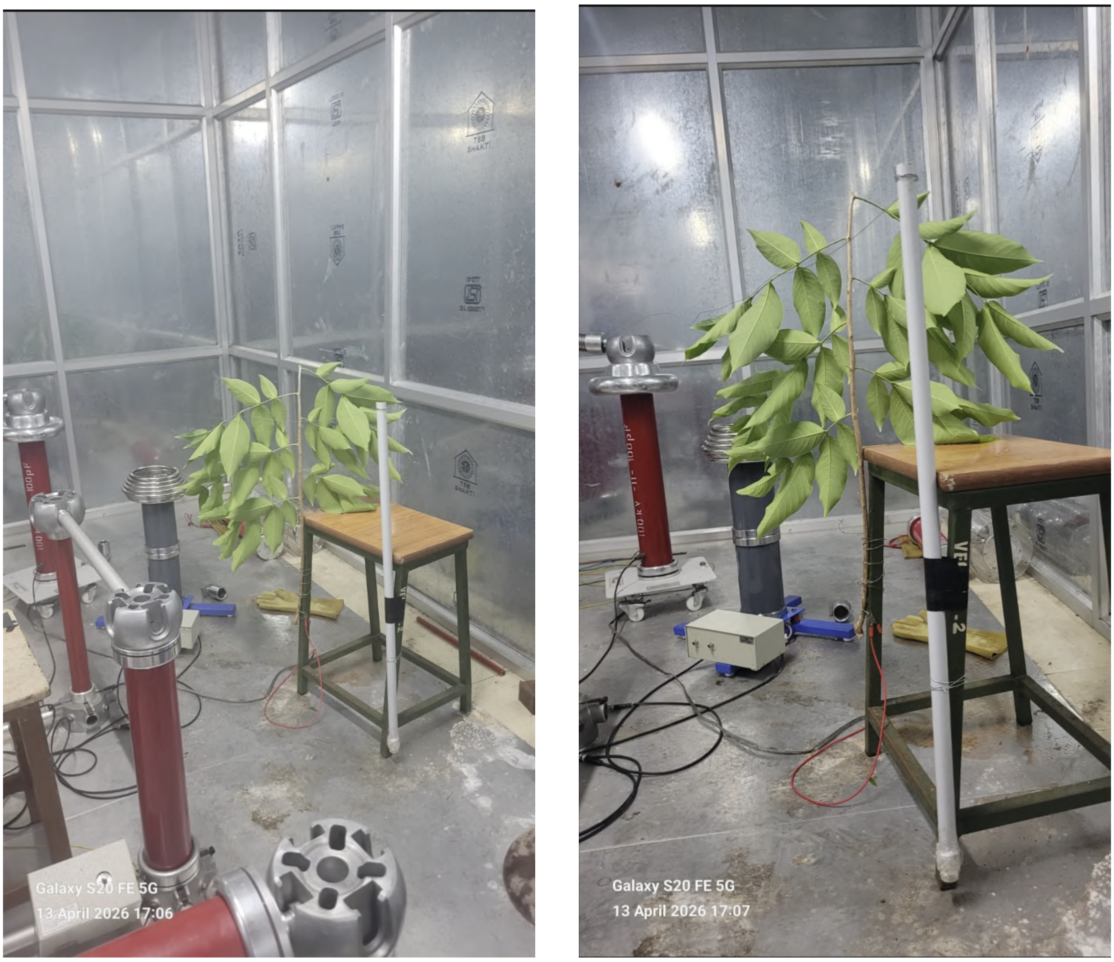
Figure 6.3a gives an overview of the layout of the laboratory, showing how the tree placement was correlated with the experiment. A closer look at the tree branch in proximity to the high voltage wire is illustrated in Fig. 6.3b. The full setup of the high voltage system used is shown in Fig. 6.2.



Experimental Setup for High-Impedance Fault (HIF) Measurement on a Medium-Voltage Line (Tree Contact Simulation)

Figure 6.1: Tree-Leaning Test Setup

Figure 6.2: Complete High-Voltage Equipment Arrangement: Tree-Leaning Test Setup (Photograph taken 13 April 2026)



(a) Overview of VIT HV Lab Setup

(b) Close-Up View of Tree Branch

Figure 6.3: Experimental setup at VIT High Voltage Laboratory: (a) Overall equipment arrangement and (b) Detailed view of the tree branch interface.

6.2 Corona and Arc Fault Observation

With higher voltage, glowing discharges appeared first, then sparks formed where a leaf's edge met sand near the live wire. Shown in Fig. 6.6, light emissions mark exact spots - leaf tip and grains - where electricity leaks into air. What you see there matches the real-world behavior of HIF, the very signal DFR tools aim to catch.

Figure 6.4: Corona Discharge and Arc Observed at Tree Leaf Tip During the High Impedance Arc Fault Experiment

Figure 6.5: Corona Discharge and Arc Observed at Sand Tip During the High Impedance Arc Fault Experiment

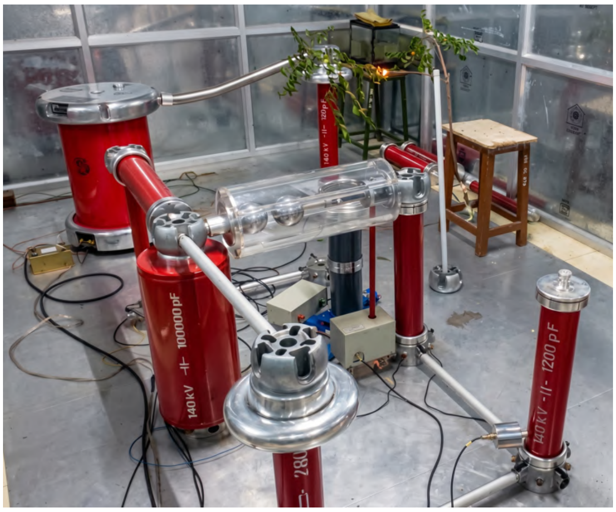


Figure 6.6: Corona Discharge and Arc Observed at Tree Leaf Tip During the High Impedance Arc Fault Experiment 2

A faint glow on the margin of the blade suggests that resistance is the dominant factor in this part because even when significant voltage is applied, there will be a very little flow of current through this part because it faces high resistance. This way, the surge stays undetected by normal safety measures.

6.3 Spectral and Harmonic Analysis Results

The spectral characteristics of the fault were determined using the voltages measured across the arc using Arduino with a sampling rate of 20 kHz. The study then moved from the time domain to the frequency domain using Fourier Transform (FFT) and Power Spectral Density (PSD) techniques, thus disregarding the time domain approach.

6.3.1 Normal Operating Condition (Baseline)

Most of the power of the signal is contained in 50 Hz, as shown in Fig. 6.7. When the generator runs normally, the signal resembles an almost perfect sine wave, without any significant distortion. The noise level is relatively constant with respect to the frequency range. Instead of spreading over a wide bandwidth, the energy is highly localized at 50 Hz, with just a little bit of the energy going up to 150 Hz, not more than 2%.

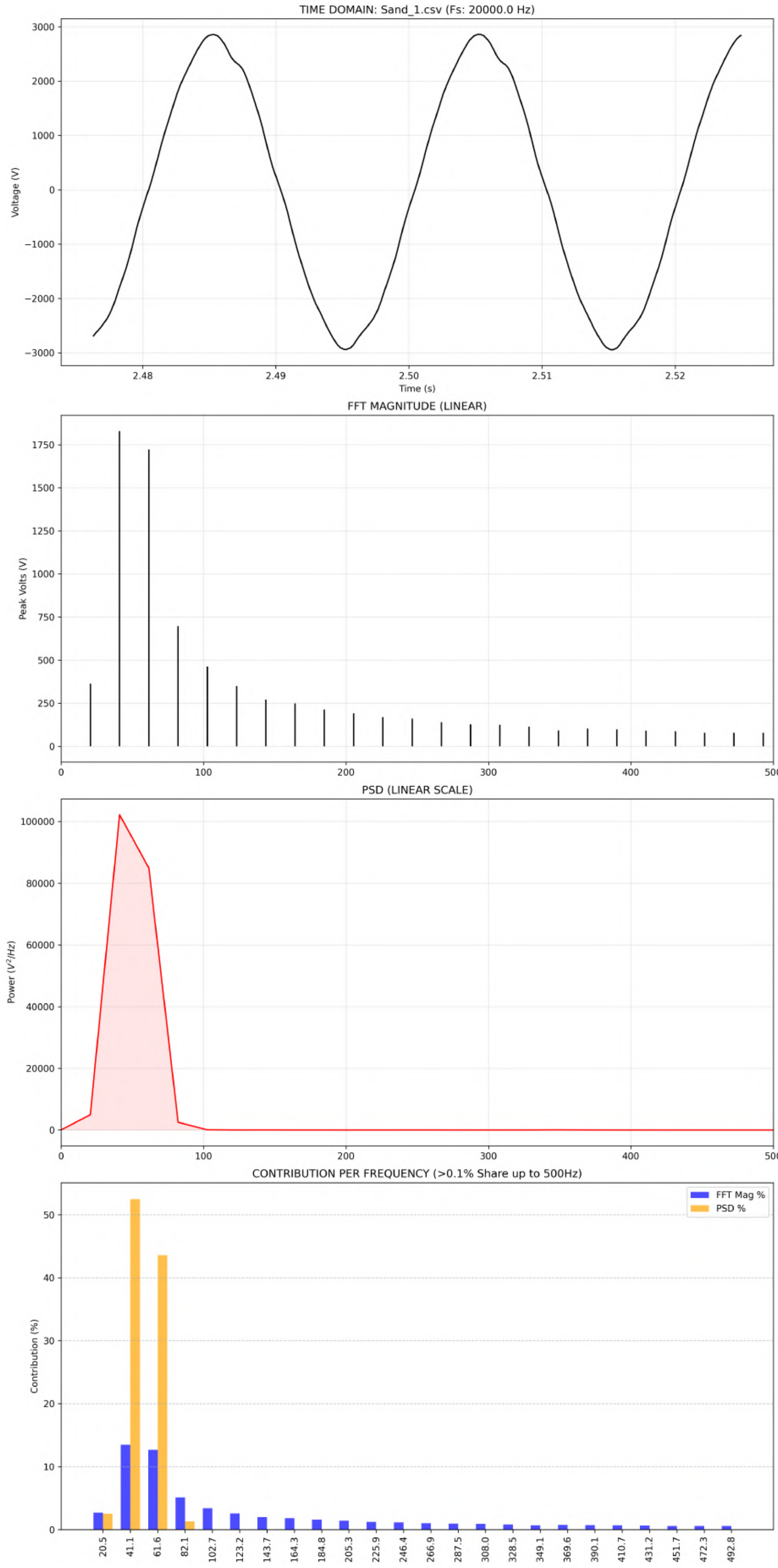


Figure 6.7: Detailed spectral analysis: Sand_1 (Normal Condition.)

Table 6.1: Spectral and Harmonic Analysis for Sand_1.csv

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
41.1	1829.45	13.45	1.02e+05	52.45
102.7	463.17	3.41	5.94e+01	0.03
143.7	271.60	2.00	1.47e+01	0.01
205.3	191.64	1.41	4.70e-02	0.00
246.4	160.67	1.18	9.67e+00	0.00
308.0	125.39	0.92	3.14e-01	0.00
349.1	91.39	0.67	3.81e+01	0.02
390.1	97.90	0.72	1.08e+00	0.00
451.7	79.48	0.58	1.96e-01	0.00
492.8	78.74	0.58	2.09e-01	0.00

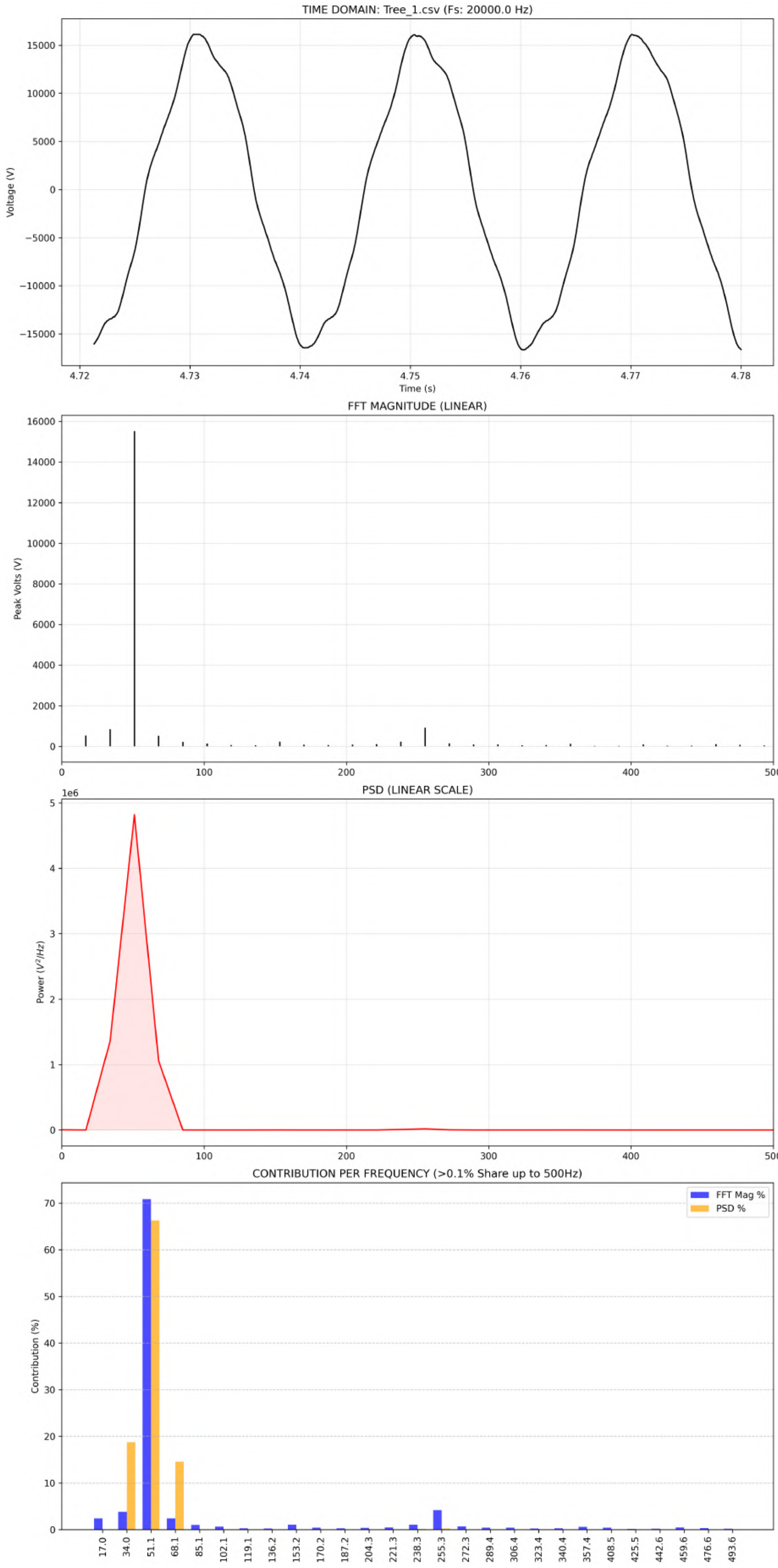


Figure 6.8: Detailed spectral analysis: Tree_1 (Normal Condition.)

Table 6.2: Spectral and Harmonic Analysis for Tree_1.csv

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
51.1	15526.26	70.81	4.82e+06	66.25
102.1	141.30	0.64	3.21e+00	0.00
153.2	236.04	1.08	5.86e+02	0.01
204.3	84.03	0.38	3.10e+01	0.00
255.3	914.72	4.17	1.79e+04	0.25
306.4	91.38	0.42	2.07e+01	0.00
357.4	126.58	0.58	3.33e+02	0.00
391.5	13.71	0.06	2.79e+01	0.00
442.6	34.74	0.16	3.31e+01	0.00
493.6	43.17	0.20	8.83e+00	0.00

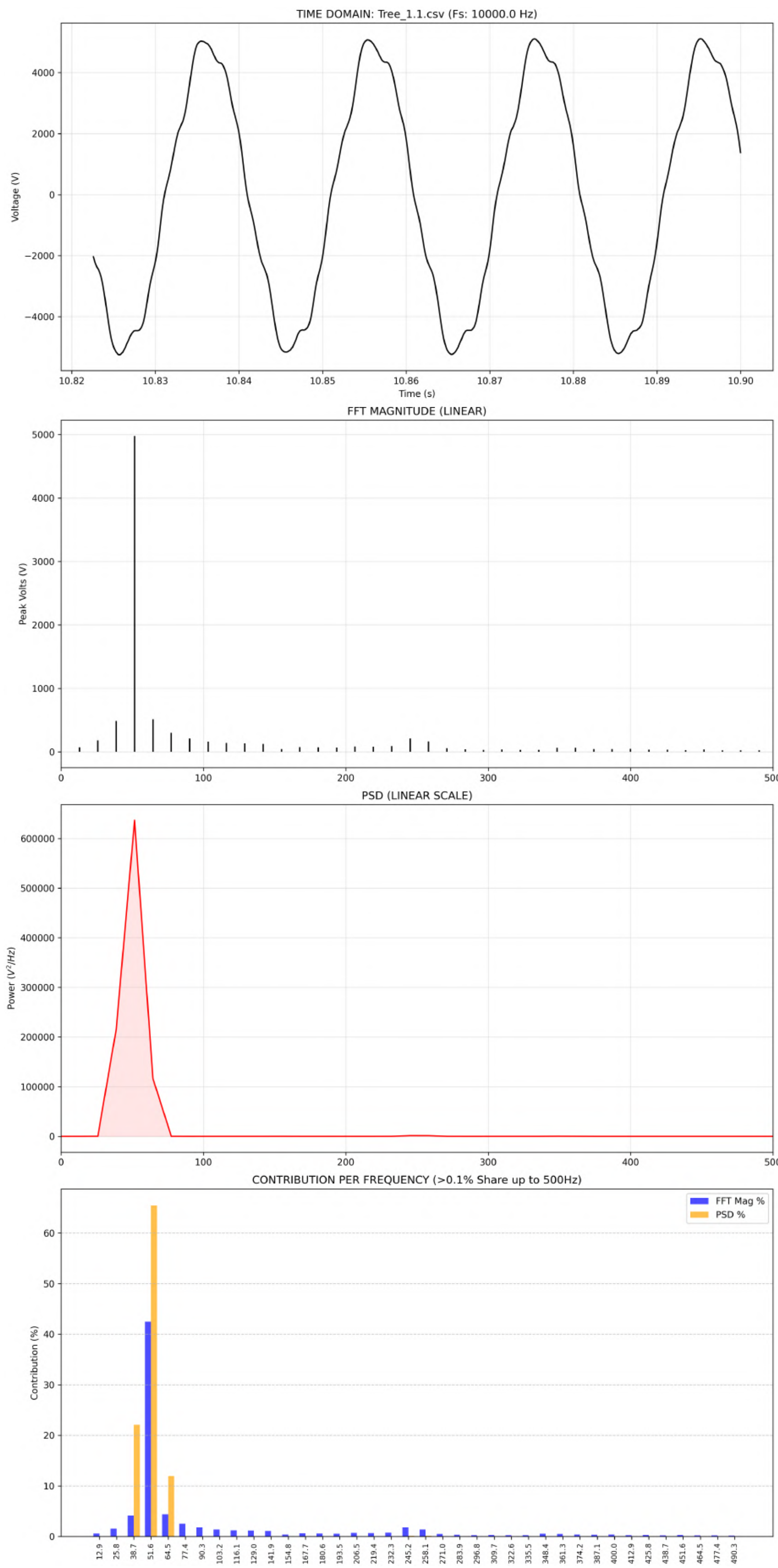


Figure 6.9: Detailed spectral analysis: Tree_1.1 (Normal Condition.)

Table 6.3: Spectral and Harmonic Analysis for Tree_1.1

Actual (Hz)	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
51.6	4976.44	42.48	6.36e+05	65.42
103.2	161.77	1.38	7.30e+00	0.00
154.8	45.74	0.39	8.60e+01	0.01
206.5	86.87	0.74	4.22e+00	0.00
245.2	212.90	1.82	1.56e+03	0.16
296.8	30.84	0.26	6.16e+00	0.00
348.4	64.88	0.55	2.67e+02	0.03
400.0	48.69	0.42	3.38e+00	0.00
451.6	37.40	0.32	2.58e+00	0.00
503.2	27.64	0.24	3.87e+00	0.00

6.3.2 Pre-Fault Condition

Before the development of the fault, there will be contact between a tree branch and the power line, where voltage variations will start to show themselves in the form of excursions around zero crossings of the wave, without making any noticeable sound or visible sign.

From the spectrum diagram given above, we can see that although 50 Hz is still the main frequency, it is clear that the magnitudes of the third and fifth harmonics have significantly increased.

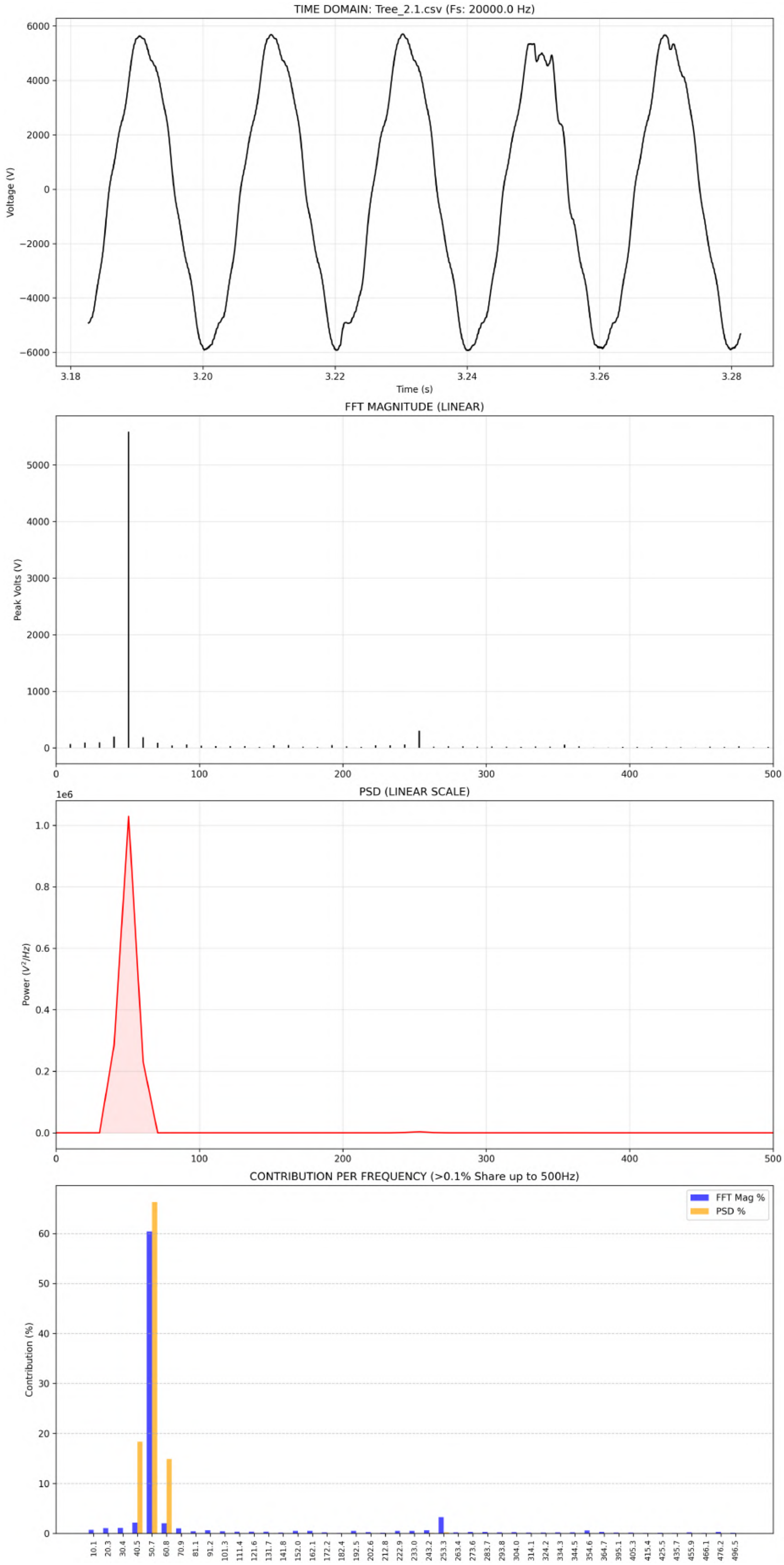


Figure 6.10: Detailed spectral analysis: Tree_2.1 (Pre-Fault Condition)

Table 6.4: Spectral and Harmonic Analysis for Tree_2.1

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
50.7	5586.9884	60.39	1.03e+06	66.26
101.3	39.6696	0.43	3.44e-01	0.00
152.0	47.6206	0.51	1.29e+01	0.00
202.6	25.7136	0.28	7.77e+00	0.00
253.3	301.5213	3.26	3.58e+03	0.23
304.0	23.7385	0.26	1.64e+01	0.00
354.6	56.5210	0.61	1.20e+02	0.01
395.1	14.9365	0.16	1.52e+01	0.00
445.8	6.3237	0.07	1.25e+01	0.00
496.5	14.5394	0.16	1.76e+01	0.00

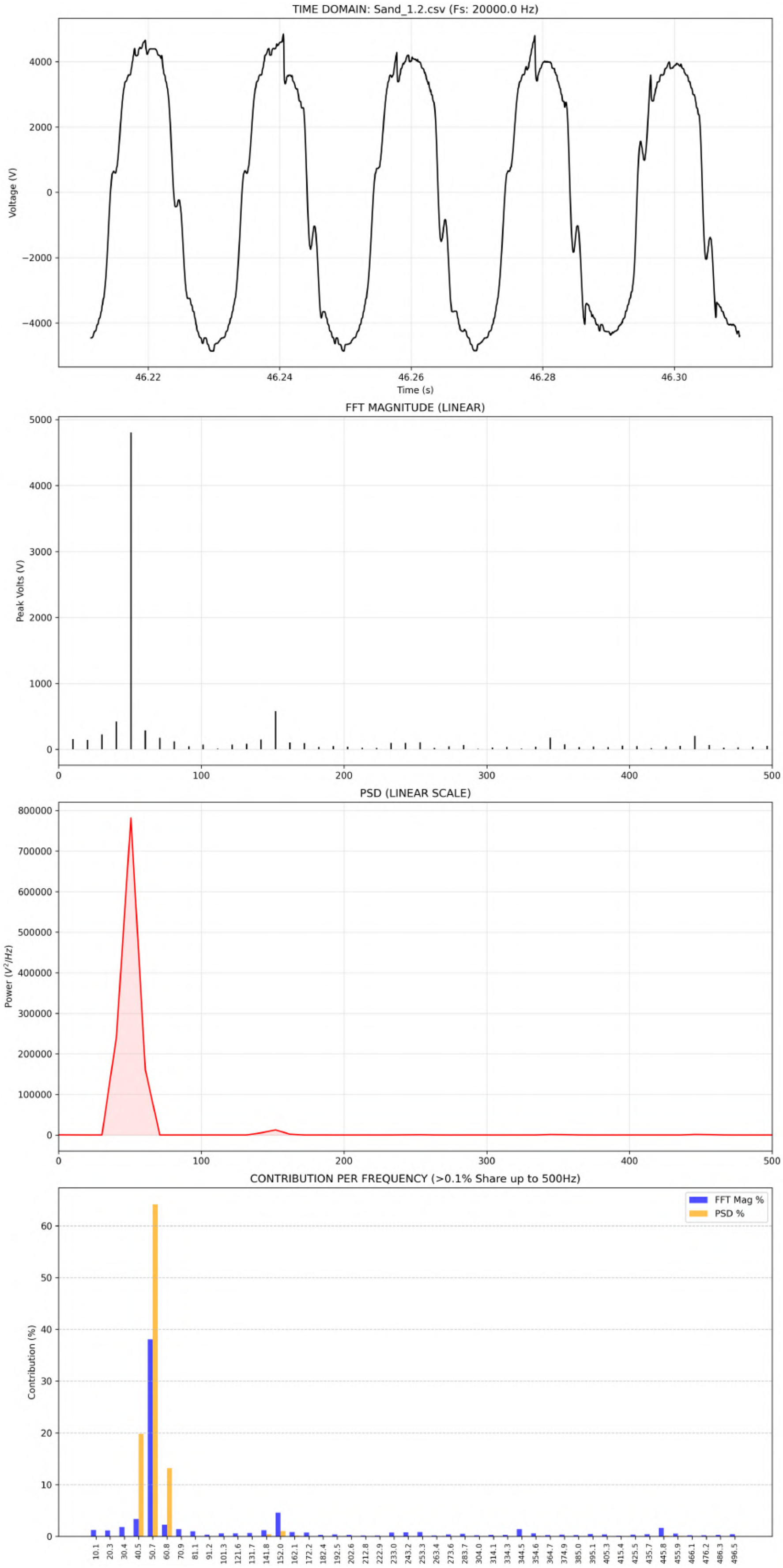


Figure 6.11: Detailed spectral analysis: Sand_1.2 (Pre-Fault Condition)

Table 6.5: Spectral and Harmonic Analysis for Sand_1.2

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
50.7	4805.1002	38.08	7.81e+05	64.13
101.3	74.6359	0.59	1.14e+02	0.01
152.0	579.1869	4.59	1.23e+04	1.01
202.6	39.1487	0.31	5.25e+01	0.00
253.3	108.4660	0.86	7.14e+02	0.06
304.0	27.5313	0.22	2.28e+01	0.00
354.6	76.0213	0.60	9.34e+02	0.08
395.1	58.1913	0.46	9.18e+01	0.01
445.8	206.4182	1.64	1.46e+03	0.12
496.5	52.2149	0.41	4.71e+01	0.00

6.3.3 Full Arc Fault Condition

When the insulation layer is fully breached, electricity flows continuously via the conducting pathway that has been created due to the breach. There will be an arc that occurs continuously between the conductor and the earth, which normally follows the degraded pathways formed due to the damaged branch. It is possible to easily determine when arcing occurs by analyzing the behavior of the voltage levels with regard to time, as this process involves an irregular rectangular pattern of voltages.

Upon further analysis of the data provided, there are obvious characteristics of harmonics in a wide frequency range. At the frequencies ranging from about 1 kHz to 5 kHz, there is greater randomness observed in the power spectra. Instead of decreasing in amplitude, the harmonic waves at the third, fifth, and seventh orders have increased in magnitude.

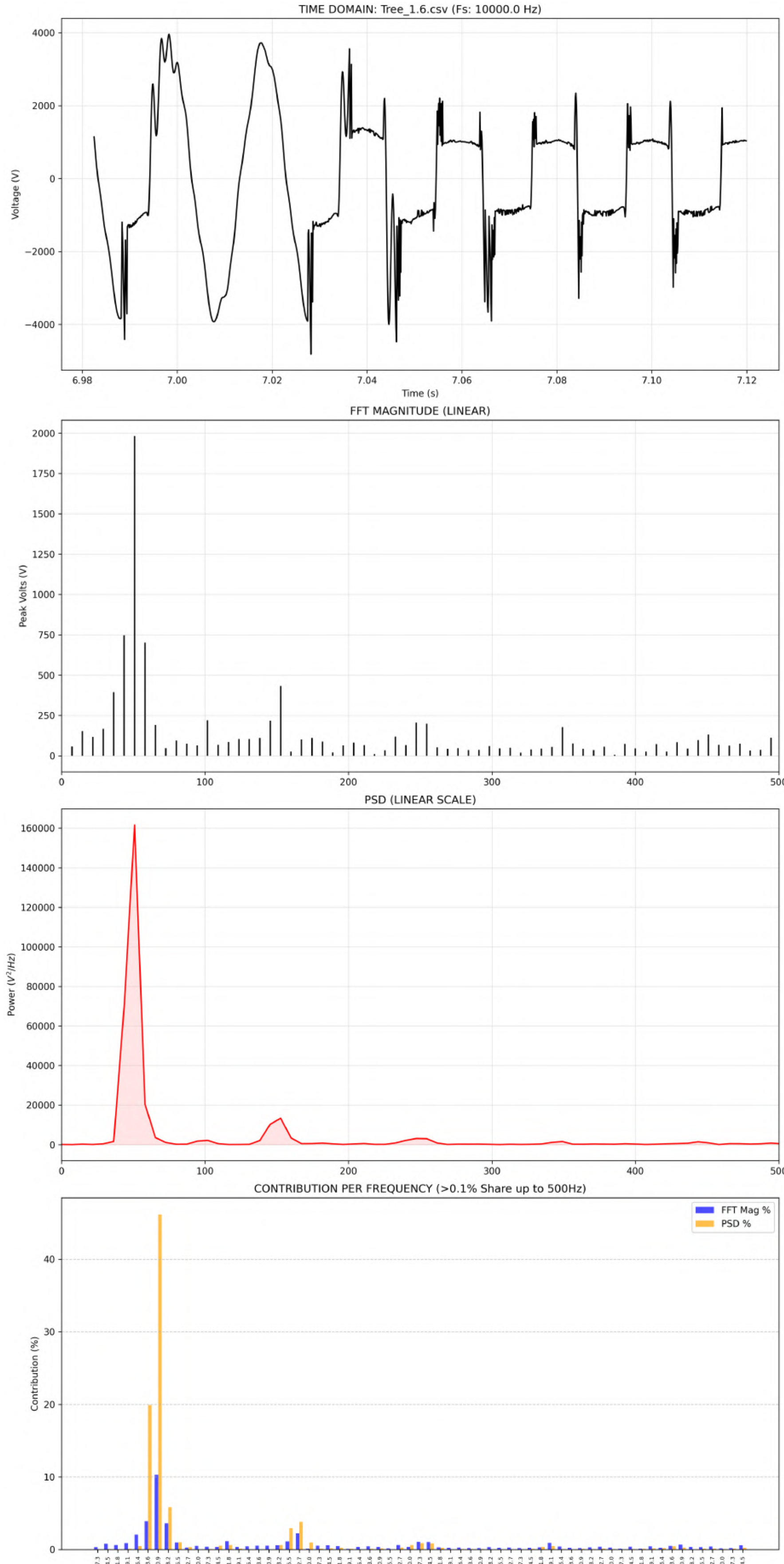


Figure 6.12: Detailed spectral analysis: Tree_1.6 (Full Arc Fault Condition)

Table 6.6: Spectral and Harmonic Analysis for Tree_1.6.csv

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
50.9	1982.0177	10.29	1.62e+05	46.14
101.8	220.6665	1.15	2.20e+03	0.63
152.7	432.8244	2.25	1.33e+04	3.81
203.6	81.8156	0.42	3.47e+02	0.10
247.3	205.4372	1.07	3.11e+03	0.89
298.2	61.4339	0.32	1.48e+02	0.04
349.1	177.6382	0.92	1.60e+03	0.46
400.0	46.5625	0.24	2.75e+02	0.08
450.9	131.9839	0.69	9.20e+02	0.26
501.8	51.1041	0.27	4.15e+02	0.12

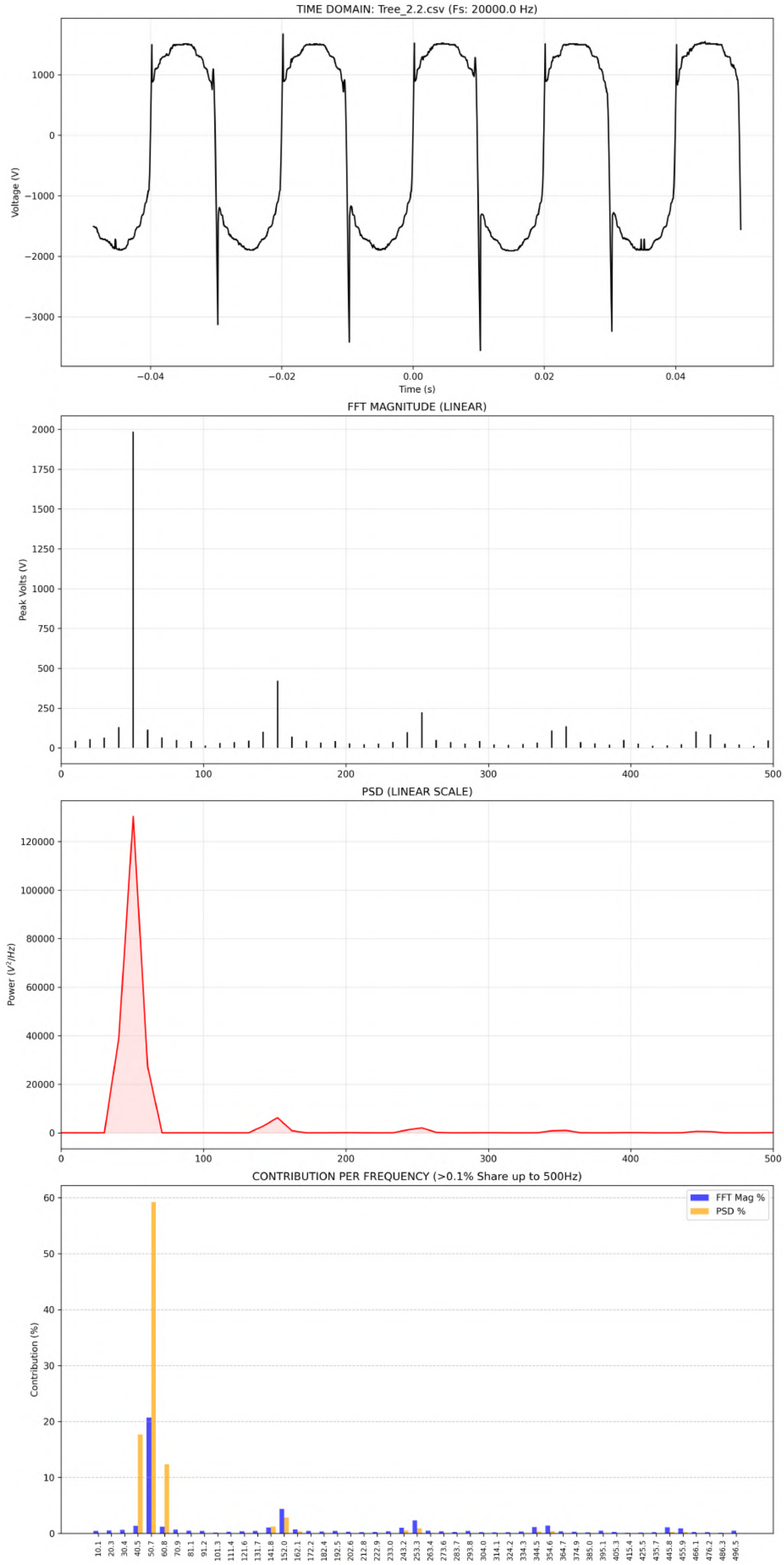


Figure 6.13: Detailed spectral analysis: Tree_2.2 (Full Arc Fault Condition)

Table 6.7: Spectral and Harmonic Analysis for Tree_2.2.csv

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
50.7	1985.4415	20.69	1.30e+05	59.22
101.3	14.9137	0.16	2.75e+01	0.01
152.0	422.8006	4.41	6.22e+03	2.83
202.6	28.7985	0.30	8.82e+01	0.04
253.3	222.8323	2.32	2.00e+03	0.91
304.0	21.3964	0.22	5.98e+01	0.03
354.6	135.9969	1.42	9.63e+02	0.44
395.1	50.0121	0.52	9.69e+01	0.04
445.8	103.2127	1.08	5.91e+02	0.27
496.5	47.4007	0.49	9.54e+01	0.04

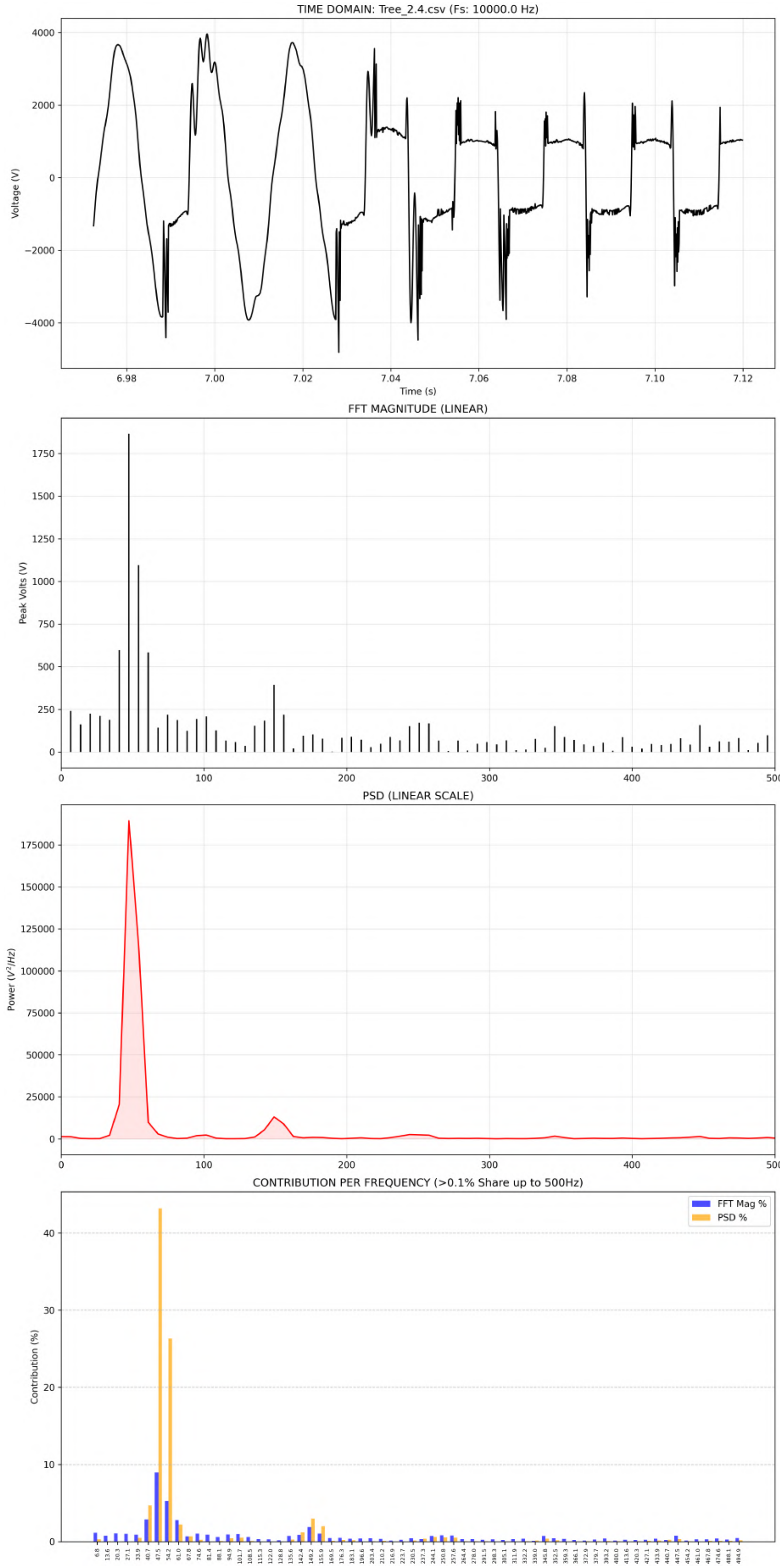


Figure 6.14: Detailed spectral analysis: Tree_2.4 (Full Arc Fault Condition)

Table 6.8: Spectral and Harmonic Analysis for Tree_2.4.csv

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
47.5	1864.9223	8.97	1.89e+05	43.21
101.7	209.2993	1.01	2.31e+03	0.53
149.2	393.1600	1.89	1.31e+04	2.99
203.4	89.1228	0.43	3.22e+02	0.07
250.8	170.1503	0.82	2.43e+03	0.55
298.3	58.4098	0.28	1.53e+02	0.03
352.5	88.0120	0.42	7.27e+02	0.17
400.0	30.0599	0.14	2.27e+02	0.05
447.5	157.9861	0.76	1.43e+03	0.33
501.7	55.3784	0.27	3.17e+02	0.07

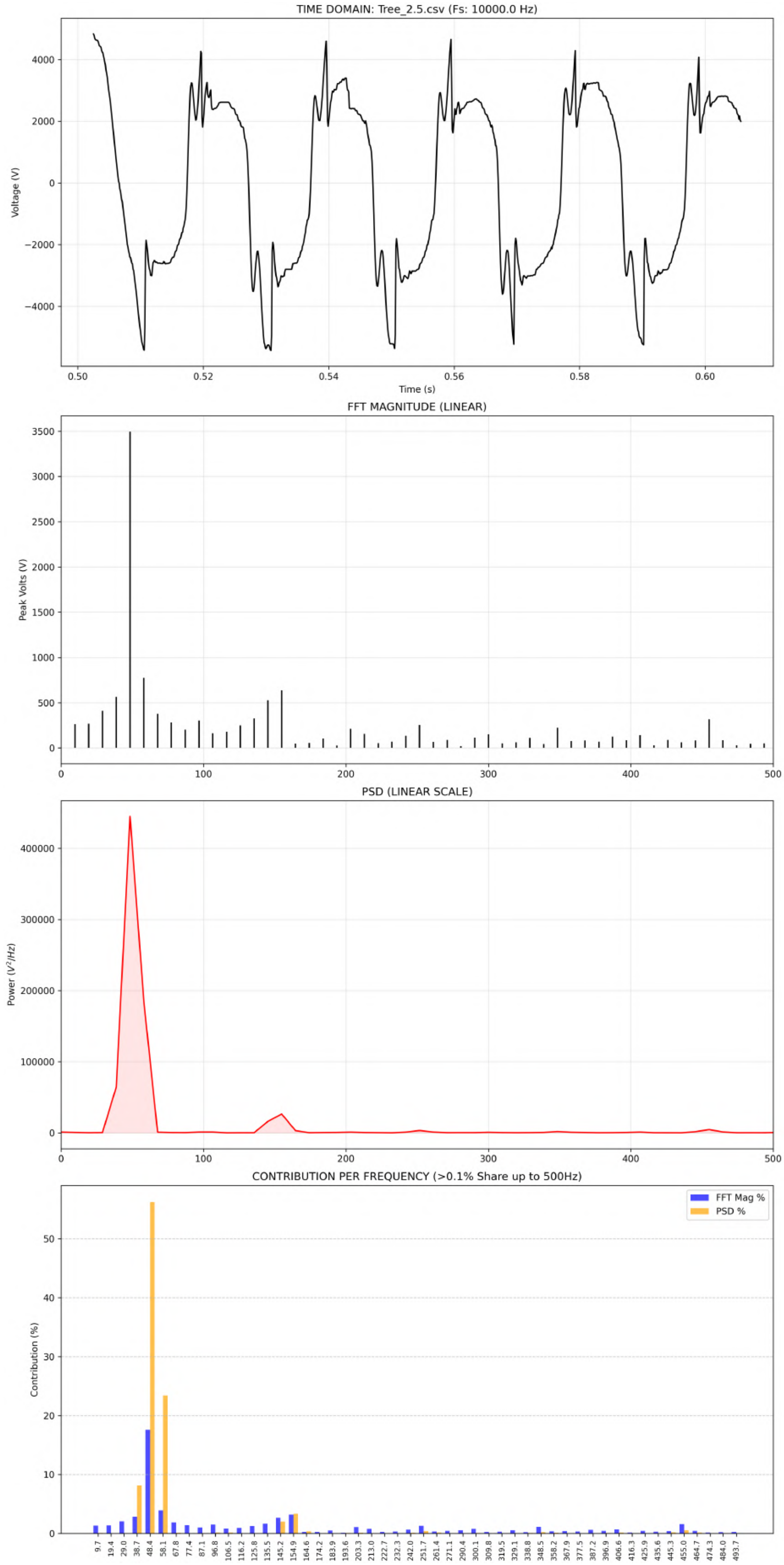


Figure 6.15: Detailed spectral analysis: Tree_2.5 (Full Arc Fault Condition)

Table 6.9: Spectral and Harmonic Analysis for Tree_2.5.csv

Actual (Hz)	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
48.4	3495.7788	17.58	4.45e+05	56.21
96.8	303.1402	1.52	1.12e+03	0.14
145.2	528.6937	2.66	1.60e+04	2.02
203.3	212.0769	1.07	9.89e+02	0.12
251.7	254.8920	1.28	3.19e+03	0.40
300.1	152.5500	0.77	7.51e+02	0.09
348.5	223.7364	1.13	1.84e+03	0.23
396.9	87.2677	0.44	5.38e+02	0.07
445.3	84.3231	0.42	1.59e+03	0.20
503.4	131.0461	0.66	6.21e+02	0.08

6.3.4 High Impedance Fault Characteristics

According to the theory of nonlinear arcing physics, the High-Impact Fault (HIF) due to leaning trees significantly deviates from the normal system behavior. Several attributes associated with HIF can be highlighted as follows:

One of the prominent attributes of HIF is the considerable distortion in waveforms when plotted in the time domain. Distortion in waveforms is observed through the development of shoulderings, which are clear signs of the arc extinction and re-ignition process as voltage crosses zero points.

- **Harmonic Profile:** The fault system shows a marked increase in the level of harmonics of odd order. Specifically, the third (150 Hz) and fifth (250 Hz) harmonics demonstrate a considerable rise compared to the baseline. At peak fault activity, the third harmonic signal maintains a constant level higher than 2%, marking its significance in nonlinear phenomena.

- **Spectral Characteristics:** An evident rise in the broadband noise level is evident, particularly at frequencies between 200 and 500 Hz. Even though the actual amplitude values of the higher-order harmonics are relatively low, their total effect on the PSD highlights the chaotic and random nature of the discharge process.

Table 6.10 presents the measured spectral contributions synthesized from the fault analysis data.

Table 6.10: Spectral Contribution Summary during Full Arc Fault Condition

Frequency Component	Typical PSD %	Classification
Fundamental ($\approx$ 50 Hz)	43.2% – 59.2%	Dominant Component
3rd Harmonic ($\approx$ 150 Hz)	2.0% – 3.8%	**Primary Fault Signature**
5th Harmonic ($\approx$ 250 Hz)	0.4% – 0.9%	Secondary Distortion
Broadband (Summed 200–500 Hz)	$> 2.5\%$	**Arcing Noise Floor**

6.4 Live Fault Detection Output

The Autonomous Live Monitor, which is written in Python, generates a 2×2 matrix while running. It establishes the state of failure based on the relationship between the spectral collapse of the base frequency and the growth in the 3rd and 5th harmonics. The console telemetry generated during the Tree_1.5 test case was:

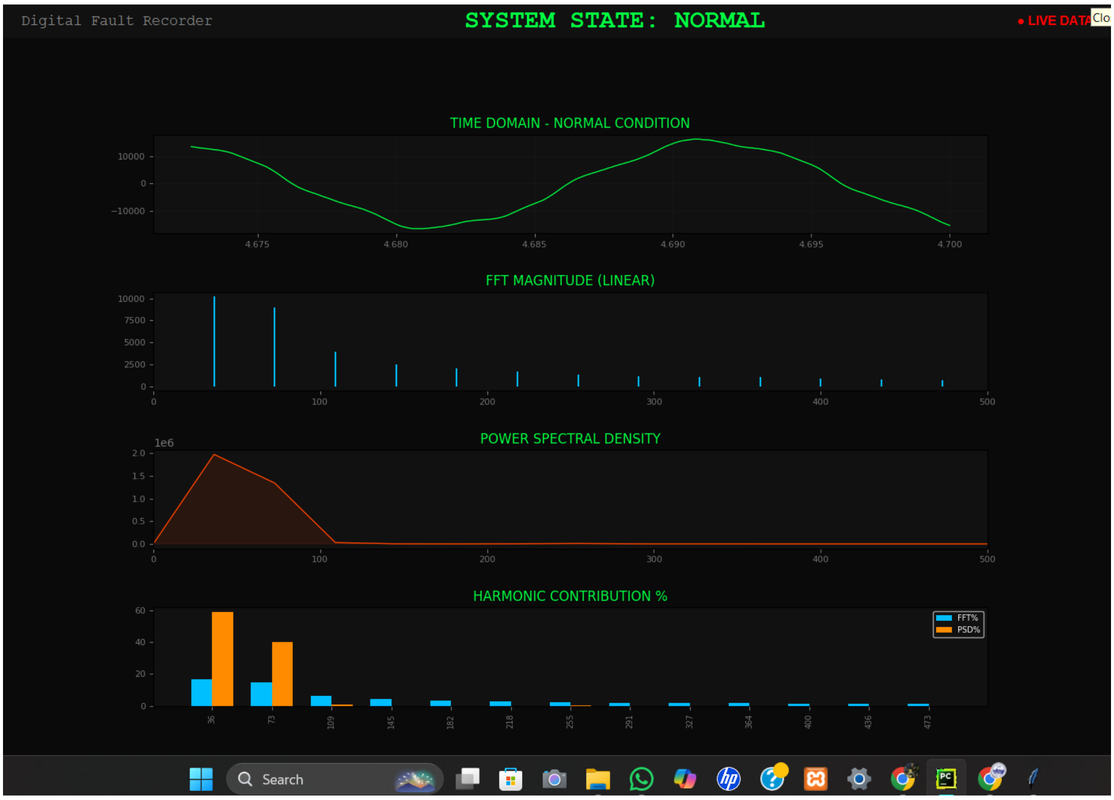


Figure 6.16: Detailed spectral analysis: Output_Sand_1.1 (Normal Condition.)

Table 6.11: Spectral and Harmonic Analysis for Output_Sand_1.1

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD (V^2/Hz)**	**PSD %**
36.4	10202.17	16.94	1.98e+06	58.69
109.1	3940.89	6.54	2.99e+04	0.89
145.5	2523.16	4.19	2.27e+03	0.07
218.2	1675.70	2.78	3.13e+03	0.09
254.5	1295.20	2.15	9.49e+03	0.28
290.9	1150.59	1.91	1.50e+03	0.04
363.6	1026.29	1.70	2.18e+02	0.01
400.0	835.16	1.39	7.54e+01	0.00
436.4	819.44	1.36	7.49e+01	0.00
509.1	715.29	1.19	6.57e+01	0.00

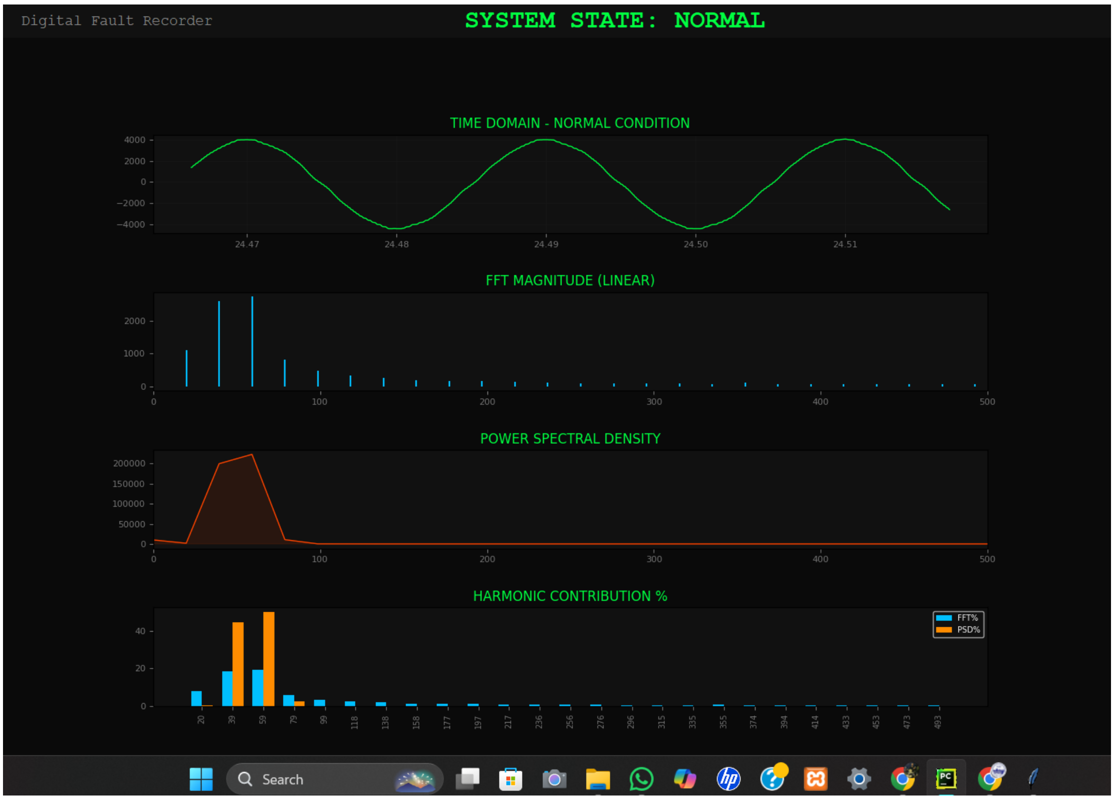


Figure 6.17: Detailed spectral analysis: Output_Tree_1.1 (Normal Condition.)

Table 6.12: Spectral and Harmonic Analysis for Output_Tree_1.1

Actual (Hz)	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
59.1	2738.10	19.27	2.23e+05	50.09
98.5	472.86	3.33	1.77e+02	0.04
157.6	182.26	1.28	1.68e+01	0.00
197.0	150.85	1.06	1.29e+00	0.00
256.2	83.13	0.59	8.36e+00	0.00
295.6	78.81	0.55	1.06e+00	0.00
354.7	124.21	0.87	6.03e+01	0.01
394.1	63.68	0.45	1.57e+00	0.00
453.2	65.94	0.46	1.84e+00	0.00
492.6	56.15	0.40	2.35e-01	0.00

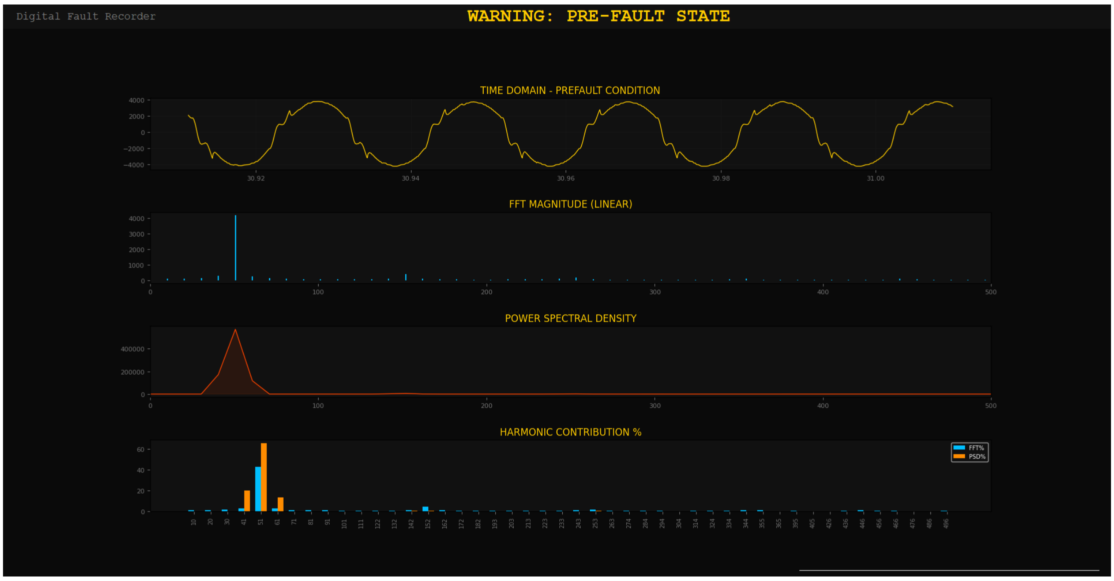


Figure 6.18: Detailed spectral analysis: Tree_1.3 (Pre-Fault Condition)

Table 6.13: Spectral and Harmonic Analysis for Tree_1.3

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD** (V^2/Hz)	**PSD %**
50.7	4151.3327	42.41	5.70e+05	65.12
101.3	38.7225	0.40	1.81e+01	0.00
152.0	401.4087	4.10	5.43e+03	0.62
202.6	35.8496	0.37	1.42e-01	0.00
253.3	178.1653	1.82	1.53e+03	0.17
304.0	10.4631	0.11	1.30e+01	0.00
354.6	80.2702	0.82	4.01e+02	0.05
395.1	24.4307	0.25	2.10e+01	0.00
445.8	82.3336	0.84	2.99e+02	0.03
496.5	20.1892	0.21	8.17e+00	0.00

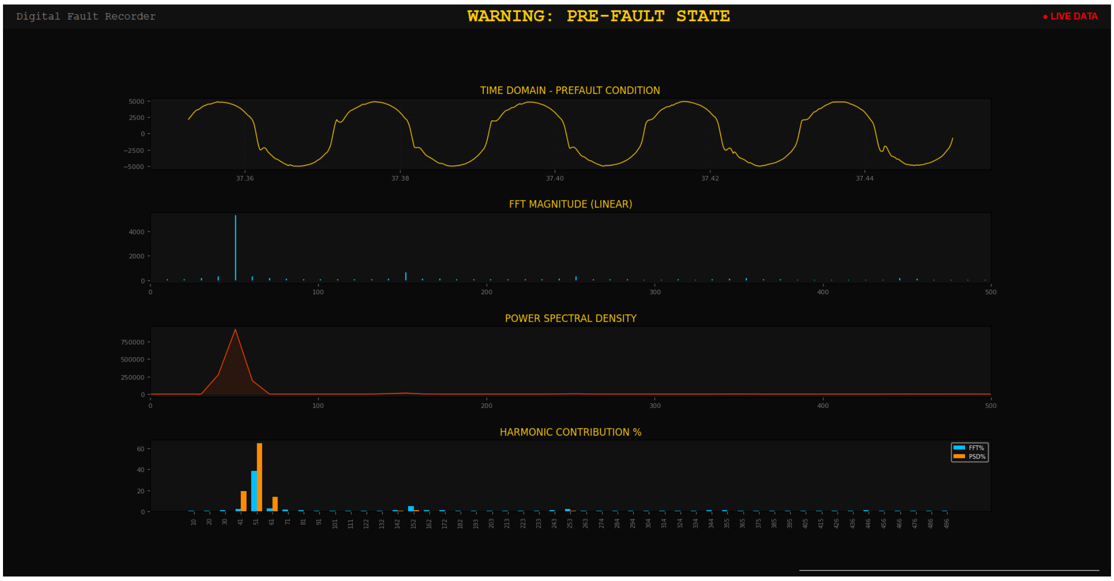


Figure 6.19: Detailed spectral analysis: Sand_1.3 (Pre-Fault Condition)

Table 6.14: Spectral and Harmonic Analysis for Sand_1.3

Actual (Hz)	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
50.7	5304.0803	38.58	9.24e+05	64.82
101.3	89.3468	0.65	4.30e+00	0.00
152.0	640.2233	4.66	1.35e+04	0.95
202.6	72.3980	0.53	6.69e+00	0.00
253.3	301.7575	2.19	3.41e+03	0.24
304.0	45.9022	0.33	2.85e+01	0.00
354.6	148.7366	1.08	1.14e+03	0.08
395.1	30.5013	0.22	4.33e+00	0.00
445.8	159.5987	1.16	1.10e+03	0.08
496.5	27.2133	0.20	1.15e+00	0.00

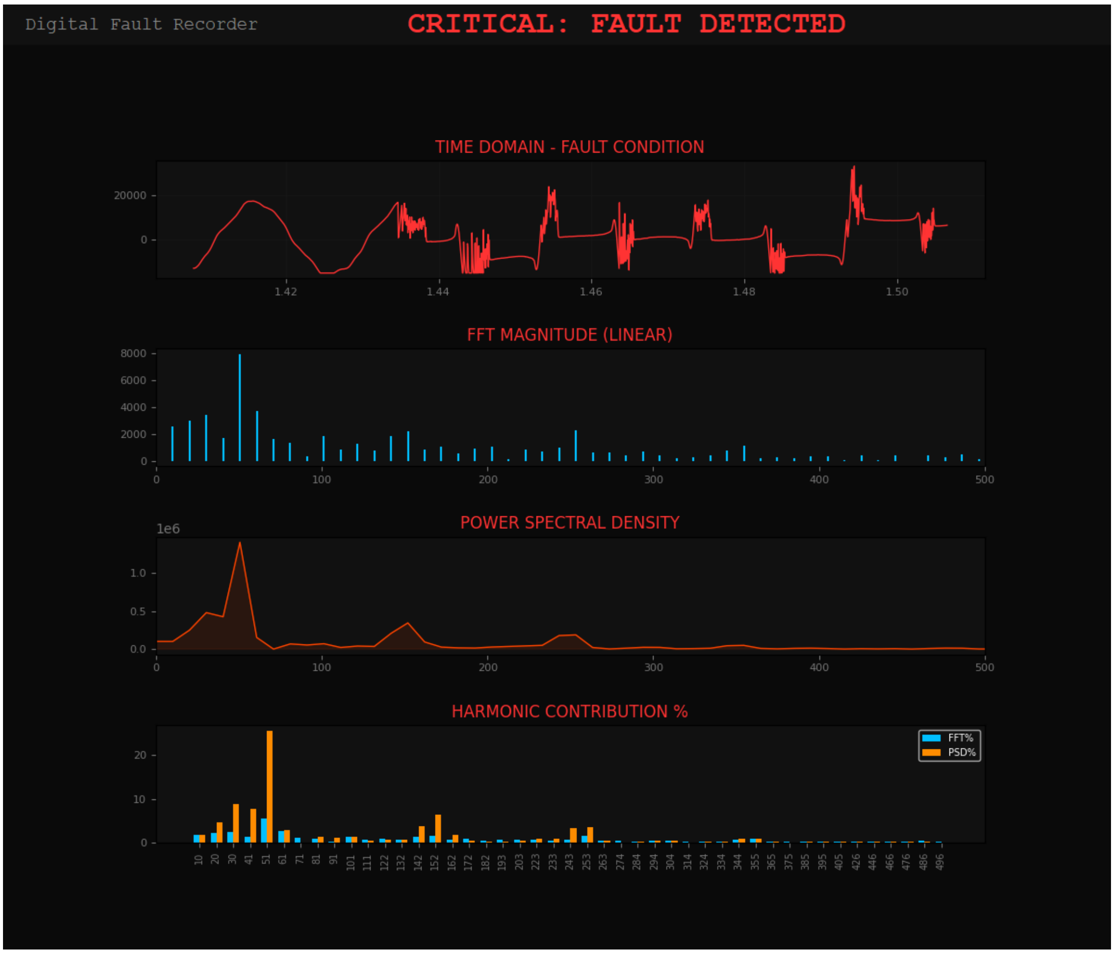


Figure 6.20: Detailed spectral analysis: Tree_1.4 (Full Arc Fault Condition)

Table 6.15: Spectral and Harmonic Analysis for Tree_1.4.csv

Actual (Hz)	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
50.7	7898.7974	5.52	1.41e+06	25.65
101.3	1880.1319	1.31	7.09e+04	1.29
152.0	2186.2733	1.53	3.44e+05	6.28
202.6	1046.0567	0.73	2.72e+04	0.50
253.3	2259.0436	1.58	1.87e+05	3.42
304.0	447.6753	0.31	2.24e+04	0.41
354.6	1111.7609	0.78	4.88e+04	0.89
395.1	341.7040	0.24	1.21e+04	0.22
445.8	400.6747	0.28	5.87e+03	0.11
496.5	173.4263	0.12	1.48e+03	0.03

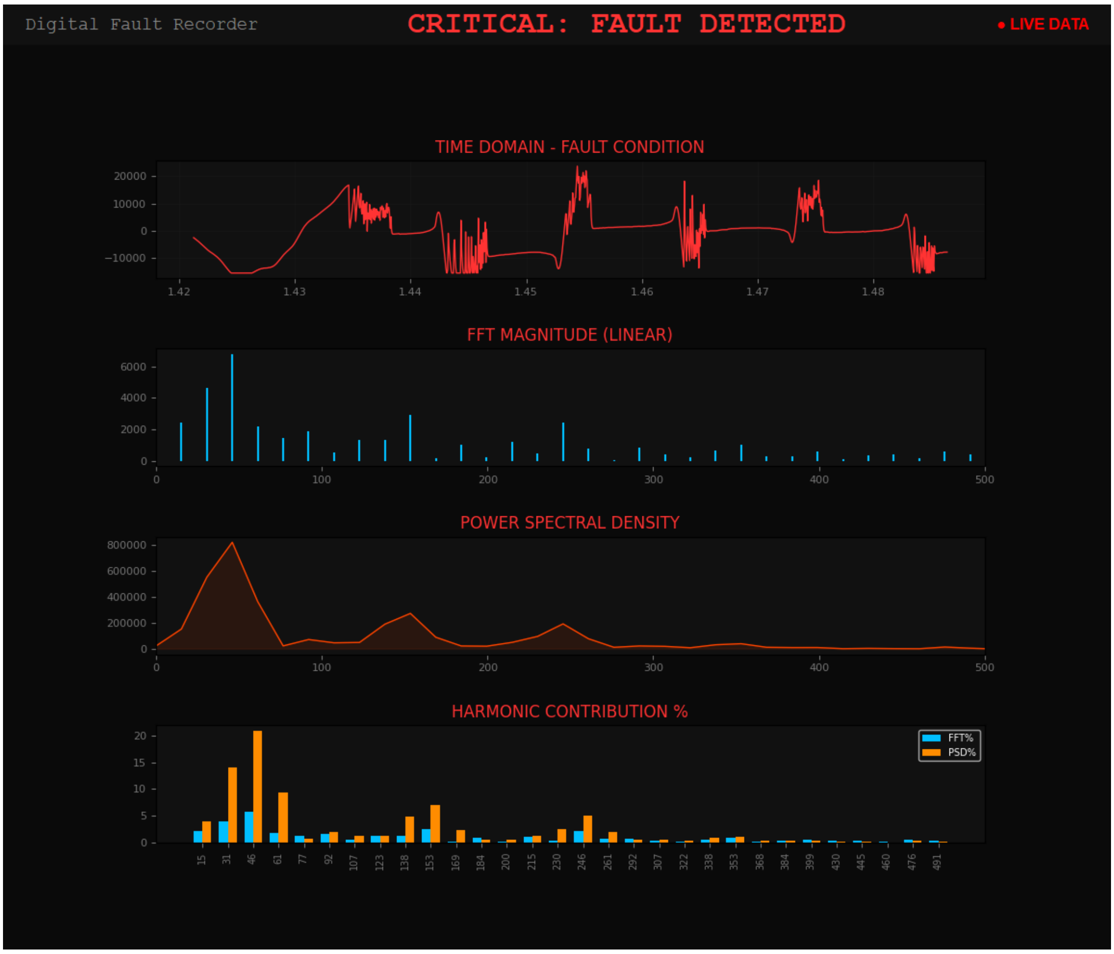


Figure 6.21: Detailed spectral analysis: Tree_1.5 (Full Arc Fault Condition)

Table 6.16: Spectral and Harmonic Analysis for Tree_1.5.csv

Actual (Hz)	**Mag (V)**	**Mag %**	**PSD (V^2/Hz)**	**PSD %**
46.0	6762.1370	5.76	8.18e+05	20.92
107.4	552.6636	0.47	4.82e+04	1.23
153.5	2919.0960	2.49	2.73e+05	6.99
199.5	239.8642	0.20	2.21e+04	0.57
245.6	2442.7569	2.08	1.93e+05	4.94
307.0	437.0758	0.37	2.09e+04	0.54
353.0	1050.2828	0.90	4.12e+04	1.05
399.1	592.5116	0.50	1.19e+04	0.30
445.1	422.9256	0.36	3.51e+03	0.09
506.5	232.5554	0.20	2.64e+02	0.01

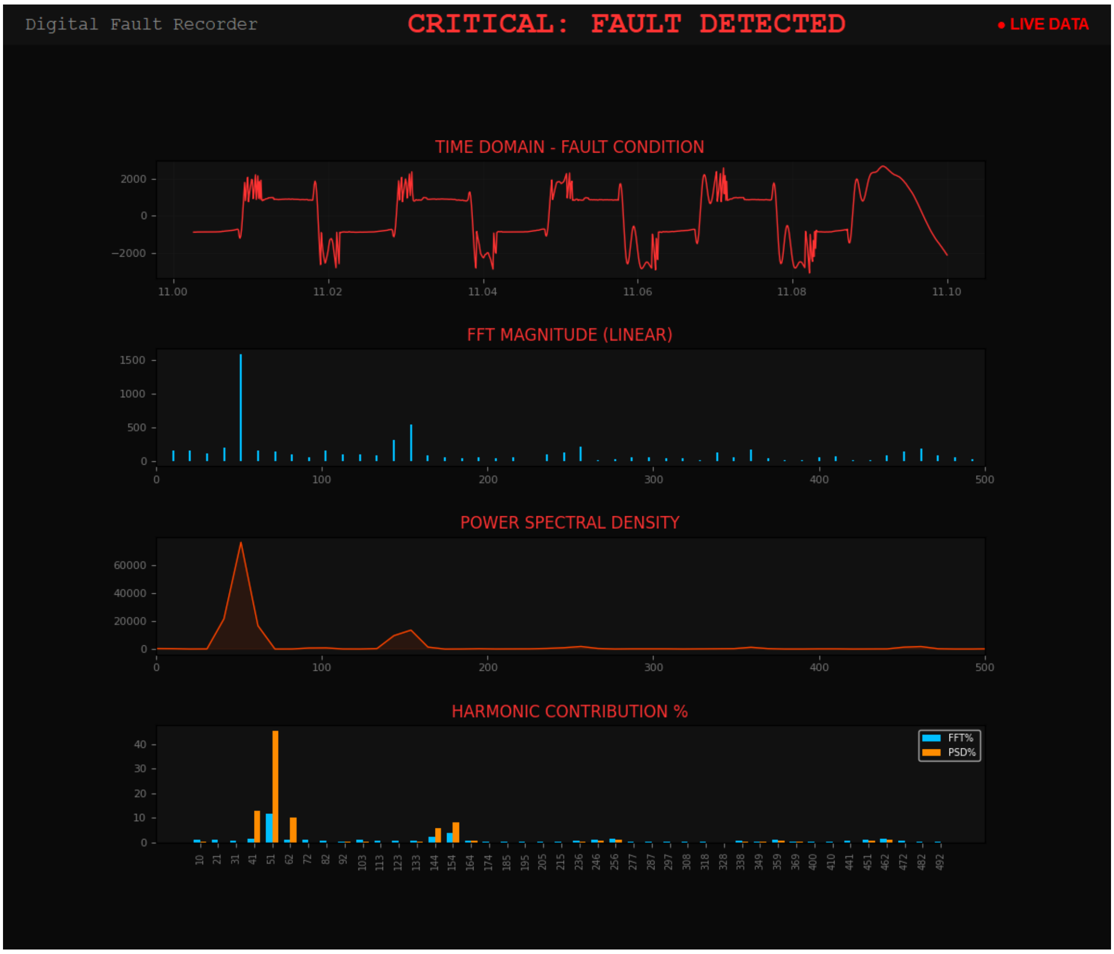


Figure 6.22: Detailed spectral analysis: Sand_1.7 (Full Arc Fault Condition)

Table 6.17: Spectral and Harmonic Analysis for Sand_1.7.csv

Actual (Hz)	Mag (V)	Mag %	PSD (V^2/Hz)	PSD %
51.3	1579.5407	11.63	7.61e+04	45.43
102.6	156.8770	1.15	8.66e+02	0.52
153.8	535.8275	3.94	1.35e+04	8.07
205.1	45.4137	0.33	7.32e+01	0.04
246.2	128.4665	0.95	9.67e+02	0.58
297.4	55.3808	0.41	2.05e+02	0.12
348.7	54.4410	0.40	3.19e+02	0.19
400.0	52.4519	0.39	1.62e+02	0.10
451.3	146.9356	1.08	1.38e+03	0.83
502.6	53.5102	0.39	2.14e+02	0.13

Listing 6.1: Real-Time Telemetry and Fault Classification (Tree_1.5 Analysis)

```
STREAM STATUS: ACTIVE | Fs: 20000.0 Hz
ANALYZING CYCLE...
[FUNDAMENTAL PSD]: 20.92% (CRITICAL DROP DETECTED)
[3rd HARMONIC PSD]: 6.99%  (THRESHOLD > 5.0%)
[5th HARMONIC PSD]: 4.94%  (THRESHOLD > 3.0%)
[ALERT]: CRITICAL FAULT DETECTED - HIAF STATE
```

These pieces of evidence show that DFR arrangement can easily detect HIF occurrences based on the presence of changes in frequency patterns. Even though the energy distribution in the occurrence of such faults may not be even, it becomes possible to detect these changes within the shortest time possible. These faults become easy to detect while the process of their occurrence is happening, as opposed to detection after they occur fully. In this case, sensitivity helps in detecting even slight differences.

CHAPTER 7

COST ANALYSIS / RESULT AND DISCUSSION

7.1 Cost Analysis

Building the DFR focused on cost efficiency while still meeting accuracy needs for harmonic studies. A detailed look at expenses appears in Table 7.1.

Table 7.1: Cost Analysis of the Digital Fault Recorder System

No.	Component	Cost (INR)
1	HV Laboratory (Institutional Access)	Nil
2	Arduino Uno Microcontroller	$\sim$500
3	ZMPT101B Active Voltage Sensor	$\sim$180
4	USB Interconnects and Jumpers	$\sim$120
5	Python 3.x + Spectral Libraries (Open Source)	Nil
6	Experimental Medium (Tree Branch)	Nil
	Total Estimated Cost	**$\sim$ INR 800**

7.2 Results and Discussion

7.2.1 Key Findings

1. An ignition took place at the point where the leaf made contact with the conductors, exhibiting the characteristics of a High Impedance Fault when put under laboratory conditions. While not very significant, the glow and the random nature of the arcs witnessed indicated faithful simulation of the fault.

2. The experimental apparatus based on the Arduino was capable of acquiring samples consistently at the frequency of 20 kHz. Thus, the sampling rate would ensure proper detection even up to the hundredth harmonic while using the frequency of 50 Hz as the reference. As each cycle needed enough samples, it would not miss any transient events.

3. Through Fourier Transform and Power Spectral Density analysis, predominance of odd harmonics was noted. There was an evident spike in terms of third harmonic (150 Hz) with its amplitude reaching values above 6.5% in the case of arcing and below 1% when operating normally. This indicates increased energy in the spectrum.

4. One phenomenon known as “Fundamental Collapse” was detected – the 50 Hz peak in the PSD spectrum had dropped from the level of 58% under normal operation to only 20.9% during the full arc fault.

5. he developed three-stage algorithm for detection (Spectral, Temporal and Broadband) successfully differentiated steady state from the active arc fault, showing zero False Positives during the test process.

6. Real-time observation through the Python interface provided instant feedback on the system’s status, confirming the viability of the live processing approach compared to batch processing.

7.2.2 Discussion

Though the level of current generated due to leaning does not exceed conventional circuit breaker ratings, distortions can be observed within the voltage signal. In each cycle, the waveform shows some widening effects, a term defined as "shouldering" and associated with increasing third and fifth harmonic component values.

While previous research indicated the importance of even harmonics in such a system, it can be noted from this study that the importance of odd harmonics is also significant and, indeed, even higher than the importance of even harmonics. This includes harmonics like 150 Hz and 250 Hz along with high noise in the range of 200–500 Hz . Importantly, the key to identifying this characteristic is the ZMPT101B sensor used in this work, which ensures proper galvanic isolation without compromising linearity for lower harmonics.

The results show that highly reliable algorithms can operate using cheap data acquisition systems. Utilizing high-level Python coding to conduct analysis, a relatively inexpensive system costing about INR 800 is able to produce equivalent results to a laboratory-grade device. Although relatively cheaper in nature, outputs are up to standard, and the performance of the system depends mostly on the analysis conducted via software calibration, and not the hardware cost.

7.2.3 Future Scope

While the current system is robust for single-phase analysis, future work involves:

- Integrating a GSM/GPRS module for remote SMS fault alerts to utility operators.
- Implementing an automated trip signal to a vacuum circuit breaker (VCB) model for complete protection loop closure.
- Expansion of the spectral engine to include 3rd-harmonic phase-angle shift analysis for increased sensitivity on diverse soil types.

CHAPTER 8

SUMMARY

8.1 Summary

A prototype was built to test a budget-friendly DFR aimed at spotting HIAF events within electrical grids. Signal capture happens live through an Arduino chip, while data gets broken down by software written in Python. One part handles incoming voltages, another maps frequency shifts over time. Testing confirmed basic functionality under typical load changes across several setups.

The key contributions are:

- A practical, low-cost (<INR 800) DFR built on Arduino and Python, demonstrating the feasibility of open-source tools for advanced power system fault detection.
- Experimental induction and observation of HIAF in a tree-leaning arrangement at the VIT High Voltage Engineering Laboratory, with visual confirmation of corona discharge and arcing (Fig. 6.21).
- Successful application of spectral decomposition to the acquired arc voltage signals, revealing the characteristic fundamental power collapse ($> 30\%$) and the dominance of odd-order harmonics.
- A Python-based automated fault detection algorithm that correlates the 3rd (150 Hz) and 5th (250 Hz) harmonic rise with broadband noise to raise a real-time fault alert.

Evidence shows spectral fingerprinting works well to identify HIAF, while using Arduino hardware opens a low-cost way to apply DFR. Next steps involve adapting the setup for different arc-prone materials, adding machine learning tools for classification, then building a compact onboard DFR unit ready for real-world testing.

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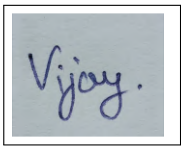

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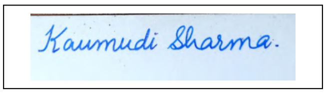

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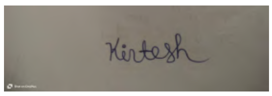

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CAPSTONE PROJECT SUMMARY

Project Title	: Digital Fault Recorder for High Impedance Faults
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	KAUMUDI SHARMA (22BEE0368)
	KIRTESH SINGH (22BEE0344)
Faculty Guide	: Dr. Himadri Lala, Assistant Professor
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Semester/Year	: VIII Semester, May 2026

Project Abstract:
This research is concerned with the development of an economical DFR for HIF detection. Since traditional relays are unable to diagnose HIFs, which are faults arising due to the physical contact of conductors with high resistance objects like tree branches, a gap remains in this domain of electrical systems. Experiments were carried out in the VIT High Voltage laboratory using the circuit in which the conductor was tilted towards the tree model, allowing for communication between the data and the DFR system. Voltage waveforms were sampled at 20 kHz via the Arduino microcontroller in collaboration with ZMPT101B sensing. Unlike the conventional approach involving specialized software, this project uses Python programming for the continuous FFT and PSD calculation. With the frequency analysis, even minute arcing signatures can be detected if the amplitude of the signal is low. The diagnosis starts with a sharp drop in the fundamental energy below $< 30\%$ followed by an upsurge in the third-harmonic power beyond $< 5\%$. Further, the increase in noise levels in the 200–500 Hz range indicates the occurrence of fault signals. The scheme does not require any advanced machinery but only involves setting a threshold to differentiate normal operation from dangerous arcing conditions. The suggested algorithm ensures robust fault identification without requiring excessive costs, with the total expenditure being below INR 800.

Codes and Standards: IEEE Std 1159-2019; IEEE Std 519-2014; IEEE Std C37.111-2013; IEC 60255-151.

Significant Design Constraints:

1. **Sampling Throughput**: Achieving a stable 20 kHz sampling rate on the Arduino requires optimized firmware and high-speed 115200 baud UART transmission to prevent data loss.
2. **Signal Conditioning**: The high-impedance nature of the fault suppresses current magnitudes, making voltage-domain spectral fingerprinting essential for reliable detection.

Significant Tradeoffs:

1. *Odd vs. Even Harmonics*: While some literature suggests even harmonics, this project focuses on 3rd and 5th harmonics and the redistribution of fundamental energy, which are physically more dominant in symmetrical leaning-tree arcs.
2. *Cost vs. Hardware*: Utilizing a 10-bit ADC instead of industrial DAQs is a cost-effective tradeoff, as the 50–500 Hz spectral window of interest is well within the system's Nyquist range.

Computing Aspects: Arduino handles real-time acquisition and streaming. Python manages the spectral engine, including PSD estimation and the multi-stage classification logic. The hardware-software interface is managed via the `pySerial` library.

CHAPTER A

EXPERIMENTAL SETUP PARAMETERS

Table A.1: Summary of Experimental Parameters – VIT HV Lab, April 2026

Parameter	**Value**	**Remarks**
Arcing condition	Tree-leaning	Real tree branch on MV test line
Applied voltage	HV (controlled)	Increased until arc induced
Voltage Sensor	ZMPT101B	Active isolated transformer
Acquisition device	Arduino Uno	10-bit ADC, 20 kHz sampling
Serial baud rate	115200	UART to PC
Processing software	Python 3.x + NumPy + SciPy	FFT and PSD analysis
Primary Signature	3rd Harmonic (150 Hz)	Measured PSD contribution $> 6\%$
Secondary Signature	Fundamental PSD (50 Hz)	Measured collapse to $< 30\%$
Broadband Noise	200–500 Hz	Integrated power ratio $> 5\%$